

1. Introduction (10 min)
Welcome to the course. Today we will discuss the importance of the immune system and how it works.

2. Immune System Overview (10 min)
The immune system is a complex network of cells and organs that work together to defend the body against pathogens.

3. Barriers to Infection (10 min)
The first line of defense is the physical barriers: skin, mucous membranes, and the lining of the respiratory and digestive tracts.

4. Innate Immunity (10 min)
This is the first response to infection. It involves cells like macrophages and neutrophils that can recognize and destroy pathogens.

5. Adaptive Immunity (10 min)
This is a more specific response that takes time to develop. It involves B cells (which produce antibodies) and T cells.

6. Memory and Vaccines (10 min)
The immune system has a memory that allows it to recognize and respond more quickly to pathogens it has encountered before. This is the basis of vaccines.

7. Immunization (10 min)
Vaccines are substances that stimulate the immune system to develop immunity to a particular disease. They are a key tool in preventing infectious diseases.

8. Immunodeficiency (10 min)
A weakened immune system can lead to increased susceptibility to infections. This can be due to genetic factors, aging, or certain medications.

9. Immunotherapy (10 min)
This is a treatment that uses the immune system to fight cancer and other diseases. It involves stimulating the immune system to attack cancer cells.

10. Conclusion (10 min)
The immune system is a remarkable defense mechanism. Understanding how it works is essential for developing effective treatments and vaccines.

11. Q&A (10 min)
We will now take questions from the audience. Please feel free to ask anything you are unsure about.

12. Summary (10 min)
We have covered the basics of the immune system, from barriers to infection to the role of memory cells. This is a complex system, but understanding its principles is crucial.

13. Next Steps (10 min)
In our next lecture, we will delve deeper into the molecular mechanisms of the immune response. We will also discuss the latest research in immunology.

14. Thank You (10 min)
Thank you for attending today's lecture. I hope you found it informative and interesting. We will see you again next week.

1. Ein Körper der Masse m gleite eine schiefe Ebene der Länge s hinunter. Die Höhe der Ebene sei h . Berechne die Geschwindigkeit v am Ende der Ebene.

2. Ein Pendel der Masse m schwinge um eine Ruhelage. Berechne die maximale Auslenkung x bei einer gegebenen mechanischen Energie E .

3. Ein Wagen der Masse M auf einer schiefe Ebene der Länge s und Höhe h wird durch eine Feder mit der Federkonstante k zurückgehalten. Berechne die Federdehnung Δx .

4. Ein Körper der Masse m werde von der Höhe h auf der Erde fallen gelassen. Berechne die kinetische Energie E_{kin} unmittelbar vor dem Auftreffen auf den Boden.

5. Ein Pendel der Masse m schwinge um eine Ruhelage. Berechne die maximale Winkelgeschwindigkeit ω bei einer gegebenen mechanischen Energie E .



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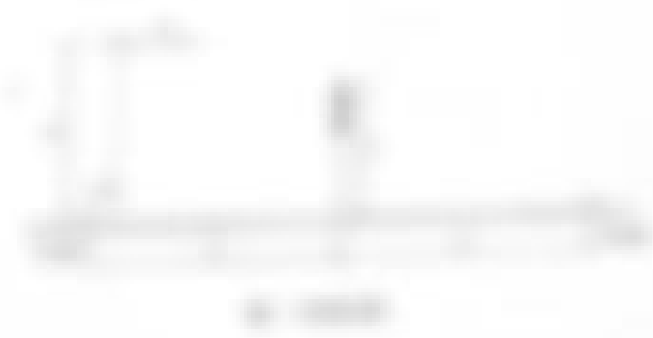
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1. **QUESTION** (10 marks)
A particle of mass m moves in a horizontal circle of radius r with constant speed v . The centripetal force is provided by a string attached to the center of the circle. The string is at an angle θ to the vertical. Find the tension in the string.

SOLUTION (10 marks)

Diagram showing a particle of mass m moving in a horizontal circle of radius r . The string is at an angle θ to the vertical. The forces acting on the particle are tension T and weight mg . The horizontal component of tension provides the centripetal force.



2. **QUESTION** (10 marks)
A particle of mass m moves in a vertical circle of radius r with constant speed v . The centripetal force is provided by a string attached to the center of the circle. The string is at an angle θ to the vertical. Find the tension in the string.

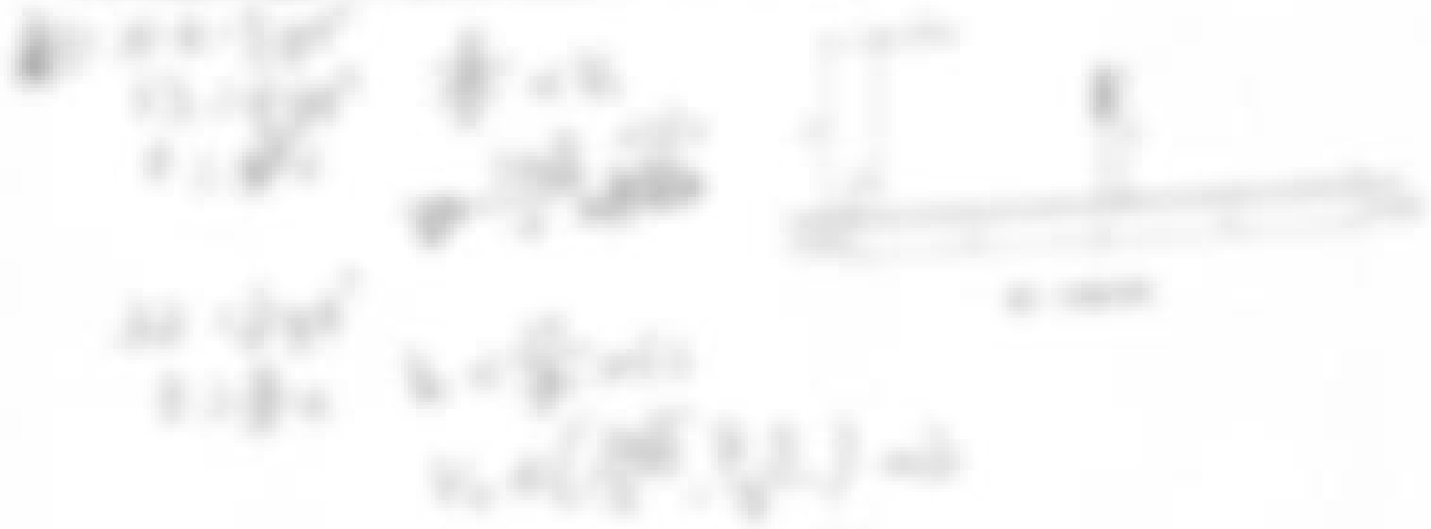
SOLUTION (10 marks)
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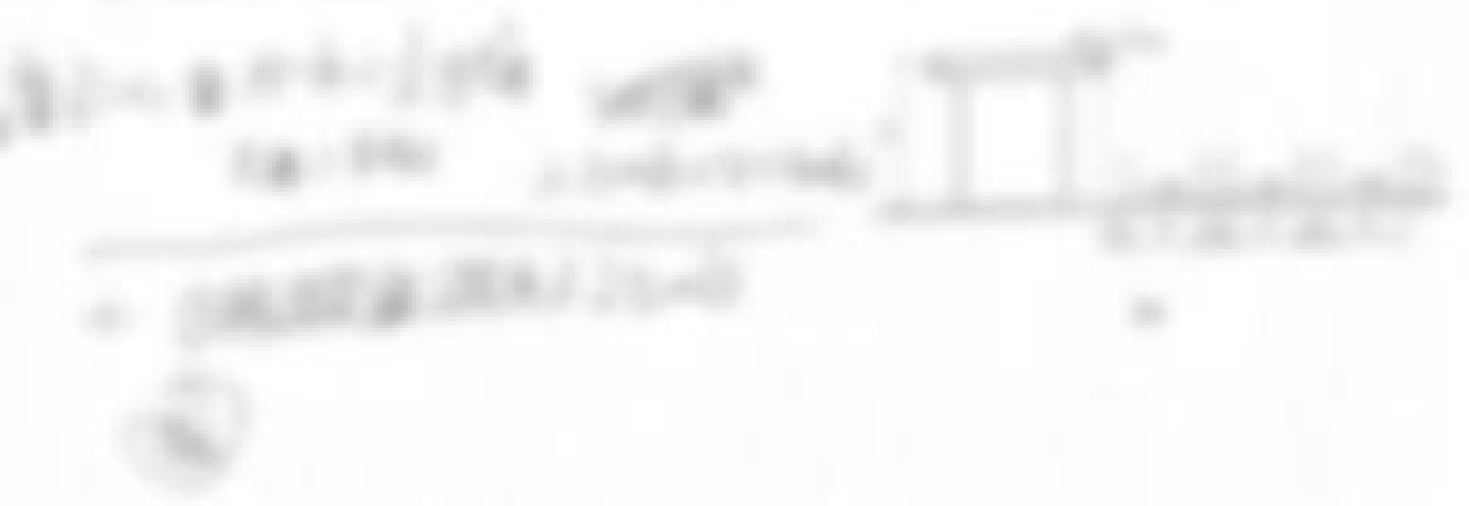
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3. Handwritten text, likely a paragraph or explanation.

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1. **QUESTION** (10 marks)
A rectangular block of material is shown in the figure below. The block is subjected to a uniform stress σ acting on its vertical faces. The block has a height h and a width b . The stress σ is acting in the positive x -direction.

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- (a) Draw a free-body diagram of the block, showing all forces acting on it.
- (b) Calculate the total force acting on the block.
- (c) Calculate the total moment acting on the block.
- (d) Calculate the total strain energy stored in the block.
- (e) Calculate the total displacement of the block.
- (f) Calculate the total work done on the block.
- (g) Calculate the total heat generated in the block.
- (h) Calculate the total mass of the block.
- (i) Calculate the total volume of the block.
- (j) Calculate the total surface area of the block.



2. **QUESTION** (10 marks)
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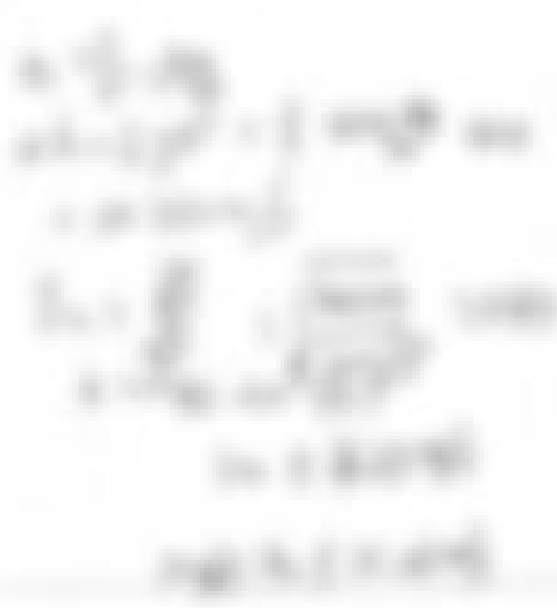
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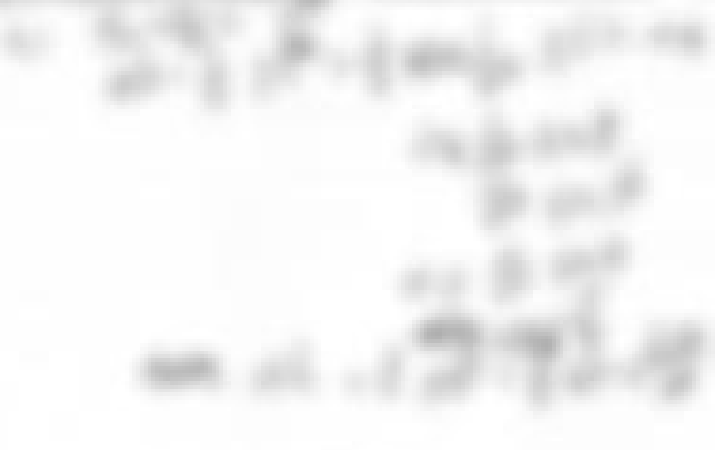
1. Perhatikan gambar berikut! Sebuah benda diletakkan pada bidang datar. Gaya berat benda bekerja ke arah bawah. Gaya normal bekerja ke arah atas. Gaya gesek bekerja ke arah kanan. Gaya tarik bekerja ke arah kiri.

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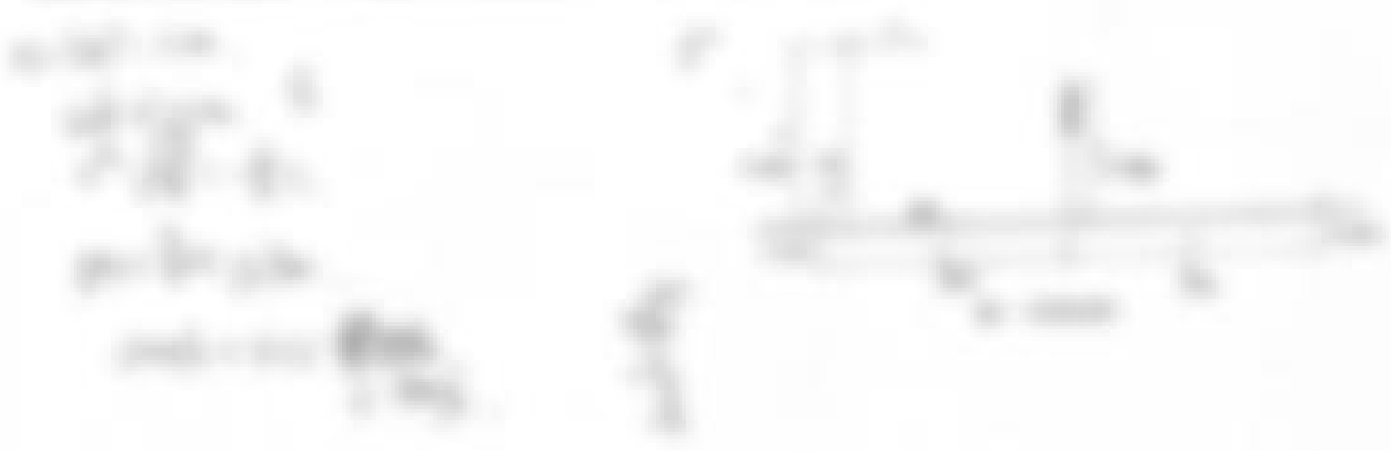
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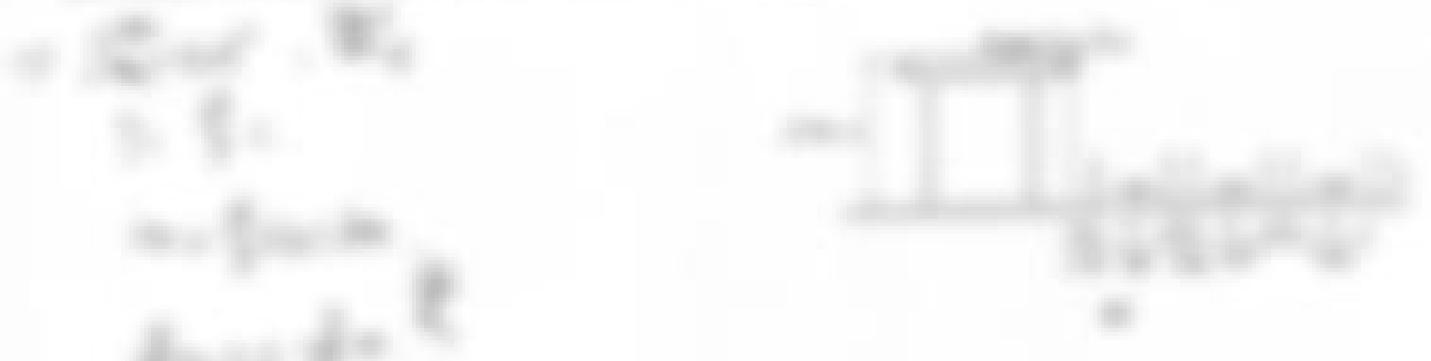
1. Handgezeichnete Schnittzeichnungen sind unverbindlich und entwerfer zu
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Schnittzeichnungen unabhängig von Maßstab und Ansicht

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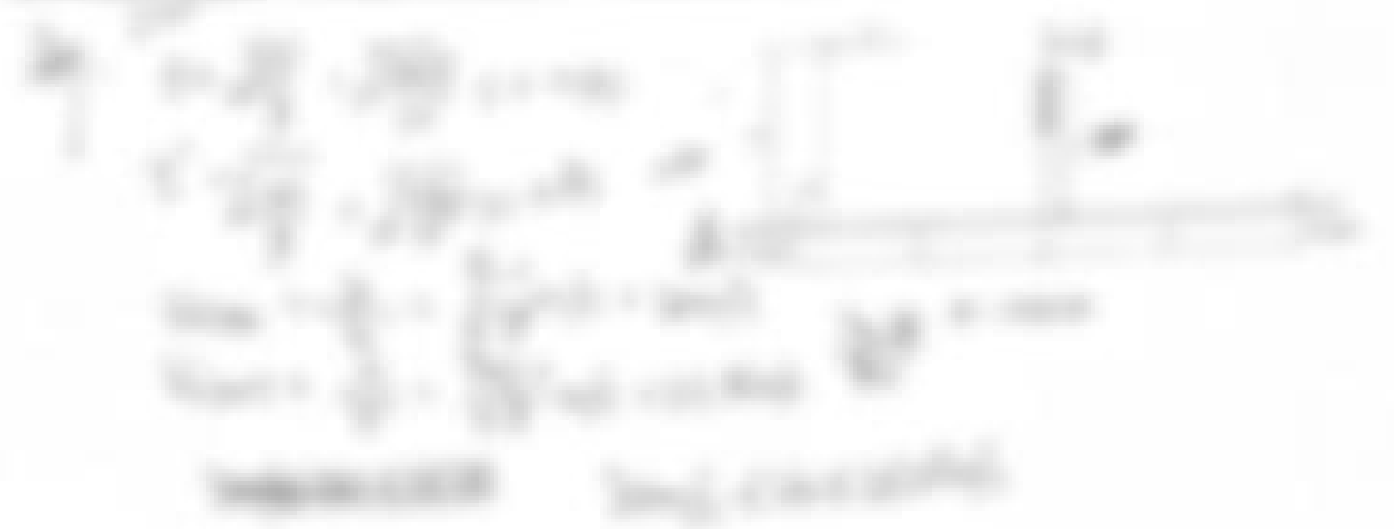
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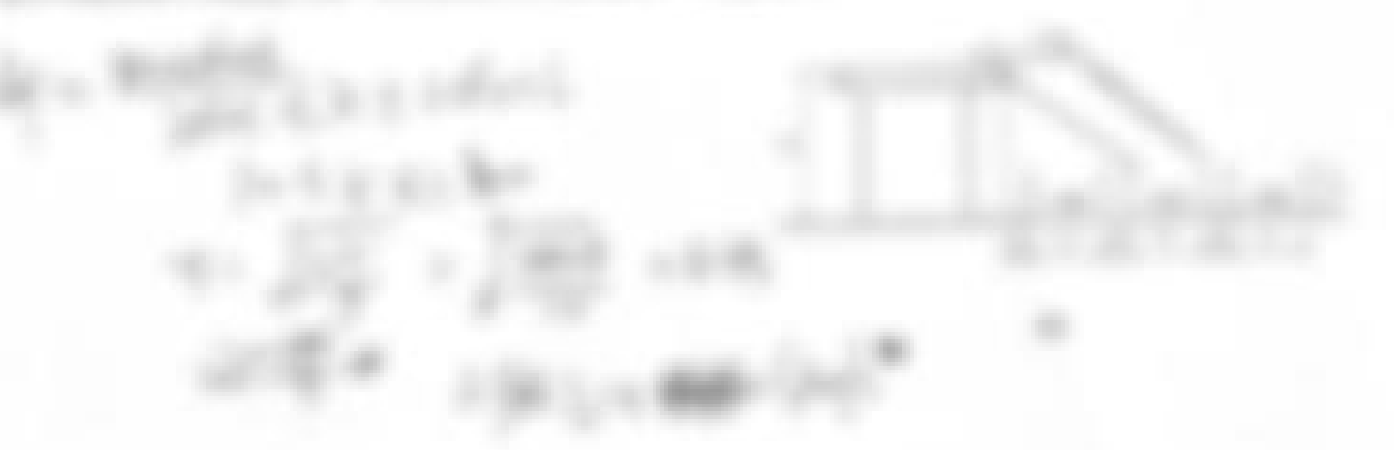
1. Geometrische Interpretation der Ableitung
Die Ableitung einer Funktion f an der Stelle x ist die Steigung der Tangente an den Punkt $(x, f(x))$ des Graphen von f .

Beispiel: Ableitung der Funktion $f(x) = x^2$ an der Stelle $x = 2$



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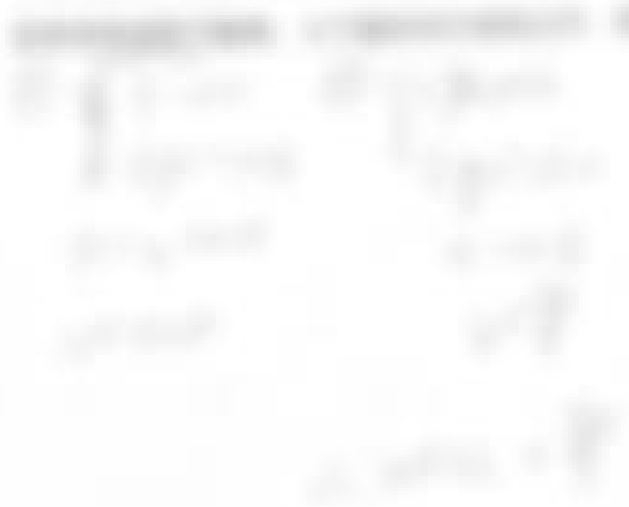
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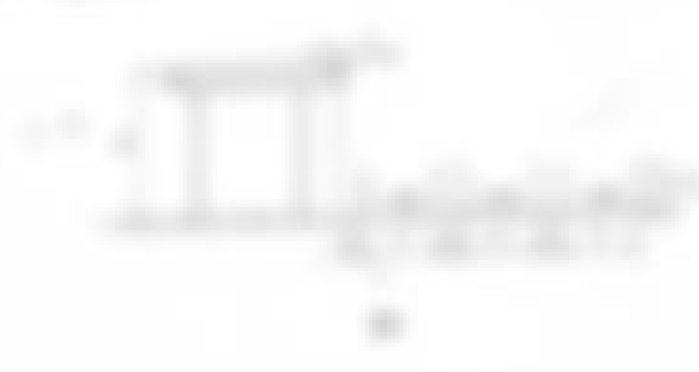


1. **QUESTION** (10 marks)
A rectangular block of mass m and height h is placed on a horizontal surface. A horizontal force F is applied to the top edge of the block, acting to the right. The block is on the verge of tipping over its right-hand edge. Find the magnitude of the force F in terms of m and h .



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1. *Journal of Management Education*, 2000, 24(1), 1-10.

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Journal of Internal Medicine 247: 395–402

2018

1. The following information relates to the operations of a company for the year ended 31st December 2018. The company has a fixed asset base of \$100,000 and a working capital of \$20,000.

Operating income: \$15,000

Operating expenses: \$10,000

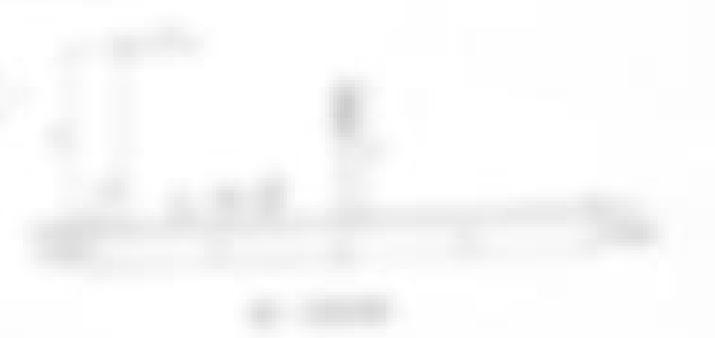
Operating profit: \$5,000

Interest income: \$1,000

Interest expense: \$500

Net profit: \$5,500

Dividend paid: \$2,000



2. The following information relates to the operations of a company for the year ended 31st December 2018. The company has a fixed asset base of \$100,000 and a working capital of \$20,000.

Operating income: \$15,000

Operating expenses: \$10,000

Operating profit: \$5,000

Interest income: \$1,000

Interest expense: \$500

Net profit: \$5,500

Dividend paid: \$2,000

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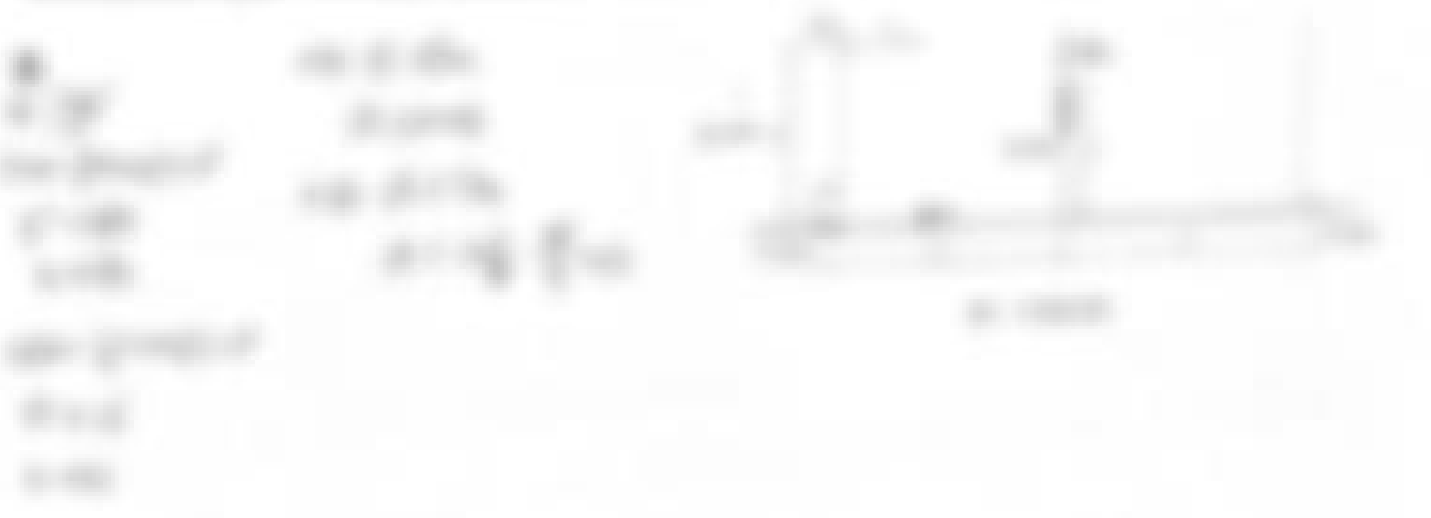
Net profit: \$5,500

U.V.



1. **QUESTION** (10 marks) The following table shows the number of students who achieved different grades in a mathematics examination. The total number of students who took the examination is 120.

TABLE 1



2. **QUESTION** (10 marks) The following table shows the number of students who achieved different grades in a mathematics examination. The total number of students who took the examination is 120.

TABLE 2

3. **QUESTION** (10 marks) The following table shows the number of students who achieved different grades in a mathematics examination. The total number of students who took the examination is 120.

TABLE 3



1. Explain the difference between a function and a procedure.
A function is a block of organized, reusable code that is used to perform a single, action. A function is simply a block of organized, reusable code that is used to perform a single, action.

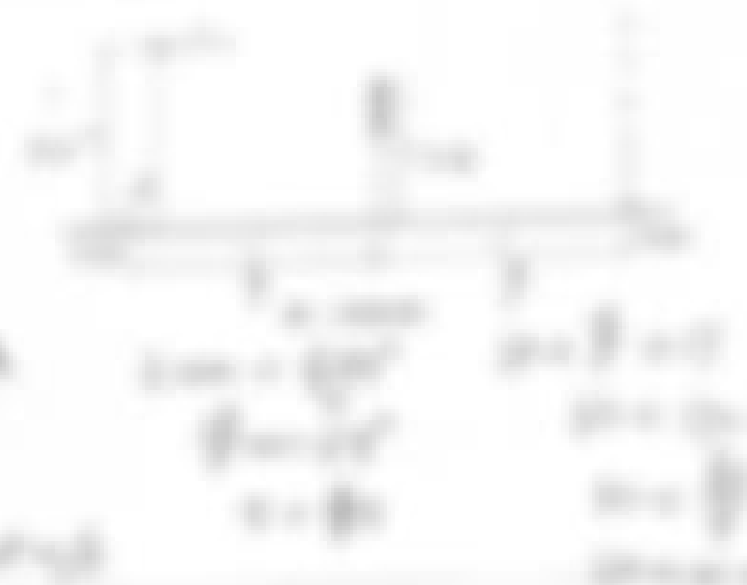
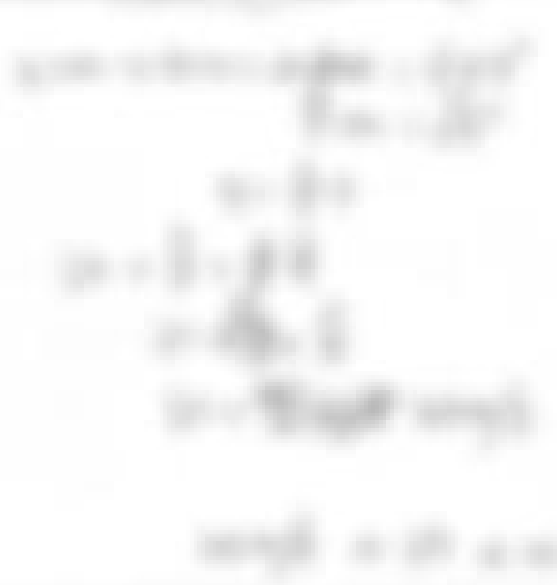
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1. Introduction (10 minutes) - Welcome to the course, overview of the syllabus, and the importance of this course in your degree.

2. Theoretical Framework (20 minutes)



2. Practical Application (30 minutes) - This section focuses on the practical application of the theoretical framework. It includes a series of exercises and case studies designed to help you understand the concepts in a real-world context.

3. Case Studies (15 minutes)

3. Case Studies (15 minutes) - This section presents several case studies that illustrate the application of the theoretical framework in different contexts. These case studies are designed to help you understand the complexities of the real world and how the theoretical framework can be used to analyze and solve problems.



1. The following information is given for the structure shown in the figure. The structure is a frame with a horizontal beam of length 10 m and a vertical column of height 4 m. The beam is supported by a pin support at the left end and a roller support at the right end. The column is supported by a pin support at the base. The beam is subjected to a uniformly distributed load of 2 kN/m acting downwards. The column is subjected to a horizontal point load of 10 kN acting to the right at the top. The structure is shown in the figure.

Draw the shear force and bending moment diagrams for the structure.

$V = 20 - 2x$



For point A

$V = 20$ and $M = 0$

For point B

$V = 18$ and $M = 20$

For point C

$V = 16$ and $M = 40$

2. The following information is given for the structure shown in the figure. The structure is a frame with a horizontal beam of length 10 m and a vertical column of height 4 m. The beam is supported by a pin support at the left end and a roller support at the right end. The column is supported by a pin support at the base. The beam is subjected to a uniformly distributed load of 2 kN/m acting downwards. The column is subjected to a horizontal point load of 10 kN acting to the right at the top. The structure is shown in the figure.

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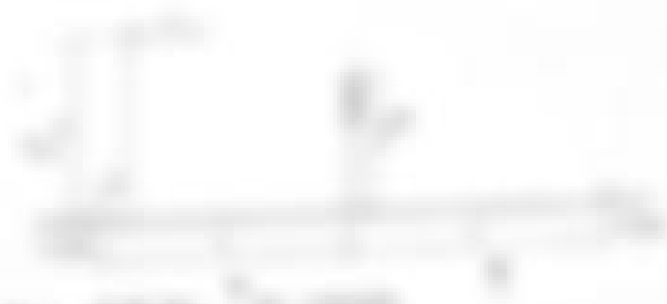
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1. A rectangular block of mass m and height h is placed on a horizontal surface. A horizontal force F is applied to the block at a height y from the bottom. The block is on the verge of tipping clockwise. The coefficient of static friction is μ_s . Find F in terms of m , h , y , and μ_s .

2. A rectangular block of mass m and height h is placed on a horizontal surface. A horizontal force F is applied to the block at a height y from the bottom. The block is on the verge of tipping clockwise. The coefficient of static friction is μ_s . Find F in terms of m , h , y , and μ_s .

- (a) $F = \frac{mgh}{2y}$
- (b) $F = \frac{mgh}{y}$
- (c) $F = \frac{mgh}{2y} + \mu_s mgh$
- (d) $F = \frac{mgh}{y} + \mu_s mgh$
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1. A horizontal beam of length 10 m is supported by a pin support at the left end and a roller support at the right end. A uniformly distributed load of 2 kN/m is applied over the entire length of the beam. Determine the reaction forces at the supports.

2. A horizontal beam of length 10 m is supported by a pin support at the left end and a roller support at the right end. A point load of 10 kN is applied at the midpoint of the beam. Determine the reaction forces at the supports.

- a) 10 kN, 0 kN
- b) 5 kN, 5 kN
- c) 10 kN, 5 kN
- d) 5 kN, 10 kN



3. A horizontal beam of length 10 m is supported by a pin support at the left end and a roller support at the right end. A uniformly distributed load of 2 kN/m is applied over the entire length of the beam. Determine the reaction forces at the supports.

4. A horizontal beam of length 10 m is supported by a pin support at the left end and a roller support at the right end. A point load of 10 kN is applied at the midpoint of the beam. Determine the reaction forces at the supports.

5. A horizontal beam of length 10 m is supported by a pin support at the left end and a roller support at the right end. A uniformly distributed load of 2 kN/m is applied over the entire length of the beam. Determine the reaction forces at the supports.

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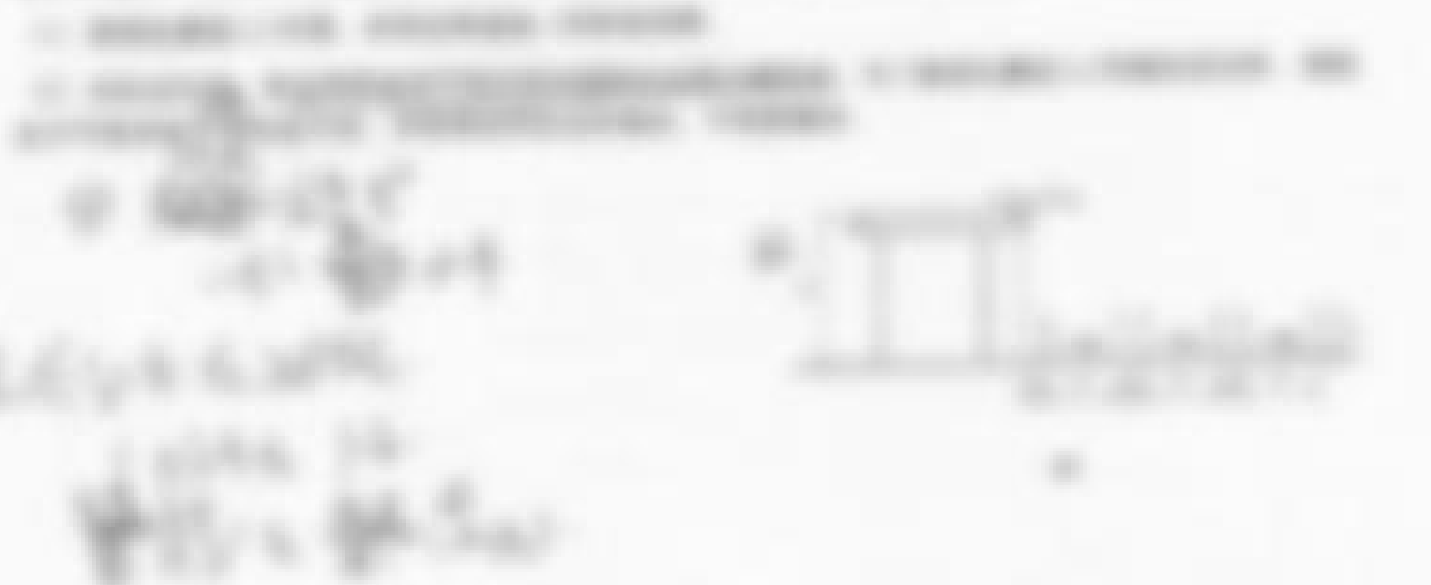


- a) 10 kN, 0 kN
- b) 5 kN, 5 kN
- c) 10 kN, 5 kN
- d) 5 kN, 10 kN

1. Introduction (10 min)
Welcome to the course. This session will cover the basic concepts of the course and the objectives of the course. We will also discuss the course structure and the assessment methods.



2. Mathematics (10 min)
This section will cover the basic concepts of mathematics, including algebra, geometry, and calculus. We will also discuss the applications of mathematics in science and engineering.



3. Physics (10 min)
This section will cover the basic concepts of physics, including mechanics, electromagnetism, and quantum mechanics. We will also discuss the applications of physics in science and engineering.



1. The first part of the diagram shows a set of nodes $U_1, U_2, \dots, U_n$ and $V_1, V_2, \dots, V_m$. These nodes are connected in a way that suggests a sequential or hierarchical relationship. The connections are represented by lines and arrows, indicating the flow of information or data between the different components.

2. The second part of the diagram shows a set of nodes $U_1, U_2, \dots, U_n$ and $V_1, V_2, \dots, V_m$. These nodes are connected in a way that suggests a sequential or hierarchical relationship. The connections are represented by lines and arrows, indicating the flow of information or data between the different components.



1. **QUESTION:** A horizontal beam of length 10 m is supported at its left end by a hinge and at its right end by a roller. A downward force of 100 N is applied to the beam at a point 4 m from the left end. Calculate the reaction forces at the supports.

SOLUTION: (a) Draw a free-body diagram of the beam.

Diagram:
A horizontal beam of length 10 m is shown. A downward force of 100 N is applied at a point 4 m from the left end. The left end is supported by a hinge, and the right end is supported by a roller.



Diagram:
A horizontal beam of length 10 m is shown. A downward force of 100 N is applied at a point 4 m from the left end. The left end is supported by a hinge, and the right end is supported by a roller.

(b) Calculate the reaction forces at the supports.

2. **QUESTION:** A horizontal beam of length 10 m is supported at its left end by a hinge and at its right end by a roller. A downward force of 100 N is applied to the beam at a point 4 m from the left end. Calculate the reaction forces at the supports.

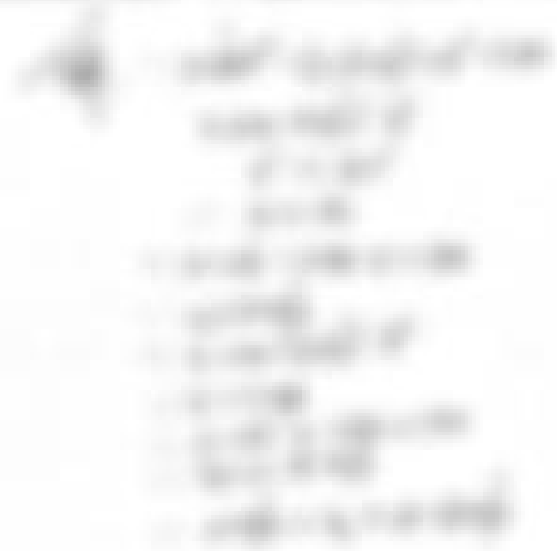
SOLUTION: (a) Draw a free-body diagram of the beam. (b) Calculate the reaction forces at the supports.

Diagram:
A horizontal beam of length 10 m is shown. A downward force of 100 N is applied at a point 4 m from the left end. The left end is supported by a hinge, and the right end is supported by a roller.

(b) Calculate the reaction forces at the supports.

1. 下列各圖中，哪一個圖形是正方形？
A. 圖 A B. 圖 B C. 圖 C D. 圖 D

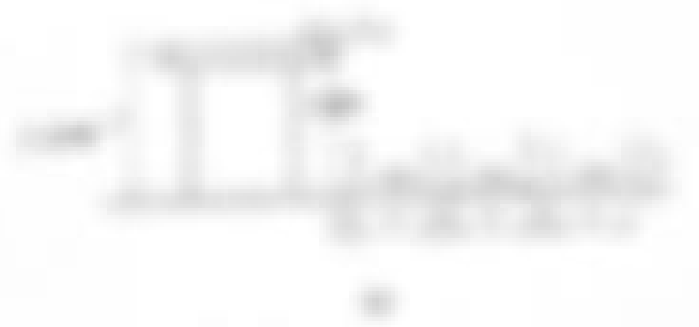
2. 下列各圖中，哪一個圖形是長方形？



3. 下列各圖中，哪一個圖形是圓？
A. 圖 A B. 圖 B C. 圖 C D. 圖 D

4. 下列各圖中，哪一個圖形是三角形？

5. 下列各圖中，哪一個圖形是四邊形？



6. 下列各圖中，哪一個圖形是五邊形？

7. 下列各圖中，哪一個圖形是六邊形？



1. The following information is given for the reaction of hydrogen gas with chlorine gas to form hydrogen chloride gas. The reaction is exothermic.

Reaction: $\text{H}_2(\text{g}) + \text{Cl}_2(\text{g}) \rightarrow 2\text{HCl}(\text{g})$

Diagram 1: A diagram showing the reaction of hydrogen and chlorine gases to form hydrogen chloride gas. It includes chemical formulas and a balanced equation.

Diagram 2: A diagram showing the reaction of hydrogen and chlorine gases to form hydrogen chloride gas. It includes chemical formulas and a balanced equation.

Diagram 3: A diagram showing the reaction of hydrogen and chlorine gases to form hydrogen chloride gas. It includes chemical formulas and a balanced equation.

2. The following information is given for the reaction of hydrogen gas with oxygen gas to form water vapor. The reaction is exothermic.

Reaction: $2\text{H}_2(\text{g}) + \text{O}_2(\text{g}) \rightarrow 2\text{H}_2\text{O}(\text{g})$

Diagram 4: A diagram showing the reaction of hydrogen and oxygen gases to form water vapor. It includes chemical formulas and a balanced equation.

Diagram 5: A diagram showing the reaction of hydrogen and oxygen gases to form water vapor. It includes chemical formulas and a balanced equation.

Diagram 6: A diagram showing the reaction of hydrogen and oxygen gases to form water vapor. It includes chemical formulas and a balanced equation.

1. A rectangular block of wood is placed on a horizontal surface. The block is 10 cm high, 20 cm wide, and 30 cm long. The weight of the block is 10 N. Calculate the pressure exerted by the block on the surface.

2. A rectangular block of wood is placed on a horizontal surface. The block is 10 cm high, 20 cm wide, and 30 cm long. The weight of the block is 10 N. Calculate the pressure exerted by the block on the surface.

$P = \frac{F}{A}$

$P = \frac{10 \text{ N}}{0.2 \text{ m} \times 0.3 \text{ m}}$

$P = \frac{10 \text{ N}}{0.06 \text{ m}^2}$

$P = 166.67 \text{ Pa}$

$P = 167 \text{ Pa}$

$P = 167 \text{ Pa}$



3. A rectangular block of wood is placed on a horizontal surface. The block is 10 cm high, 20 cm wide, and 30 cm long. The weight of the block is 10 N. Calculate the pressure exerted by the block on the surface.

4. A rectangular block of wood is placed on a horizontal surface. The block is 10 cm high, 20 cm wide, and 30 cm long. The weight of the block is 10 N. Calculate the pressure exerted by the block on the surface.

5. A rectangular block of wood is placed on a horizontal surface. The block is 10 cm high, 20 cm wide, and 30 cm long. The weight of the block is 10 N. Calculate the pressure exerted by the block on the surface.

$P = \frac{F}{A}$



1. $\frac{1}{x^2} = x^{-2}$
2. $\frac{d}{dx} x^{-2} = -2x^{-3}$
3. $= -2x^{-3}$
4. $= -\frac{2}{x^3}$

5. $\frac{d}{dx} \frac{1}{x^2} = -\frac{2}{x^3}$

6. $\frac{d}{dx} x^{-2} = -2x^{-3}$
7. $= -2x^{-3}$
8. $= -\frac{2}{x^3}$
9. $= -\frac{2}{x^3}$
10. $= -\frac{2}{x^3}$



... $\frac{d}{dx} \frac{1}{x^2} = -\frac{2}{x^3}$

1. $\frac{d}{dx} \frac{1}{x^2} = -\frac{2}{x^3}$
2. $\frac{d}{dx} x^{-2} = -2x^{-3}$
3. $= -2x^{-3}$
4. $= -\frac{2}{x^3}$

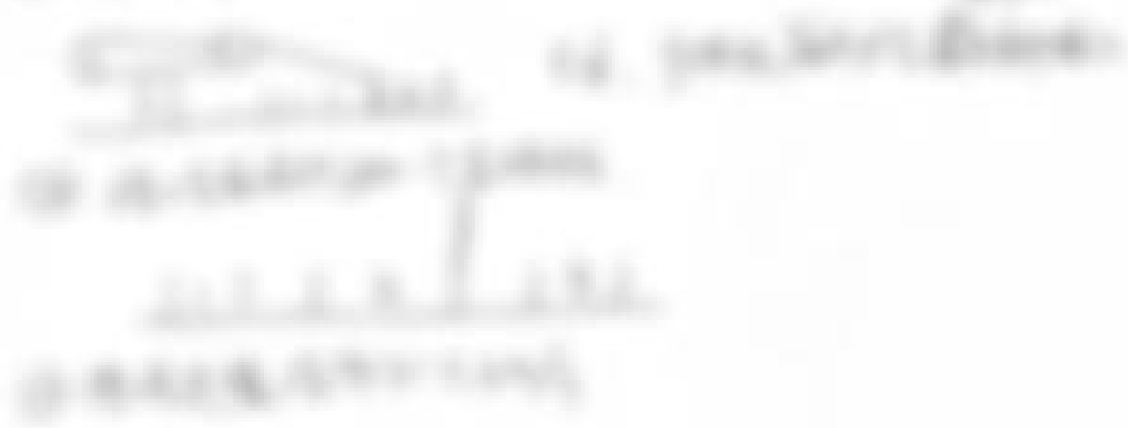
5. $\frac{d}{dx} \frac{1}{x^2} = -\frac{2}{x^3}$

6. $\frac{d}{dx} x^{-2} = -2x^{-3}$
7. $= -2x^{-3}$
8. $= -\frac{2}{x^3}$

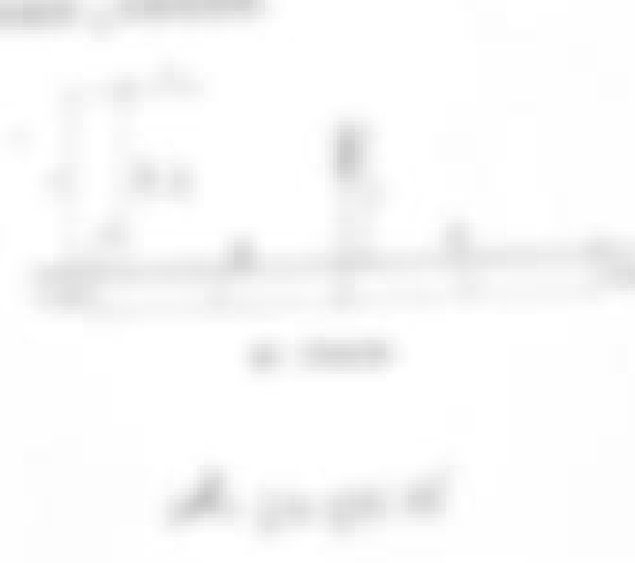
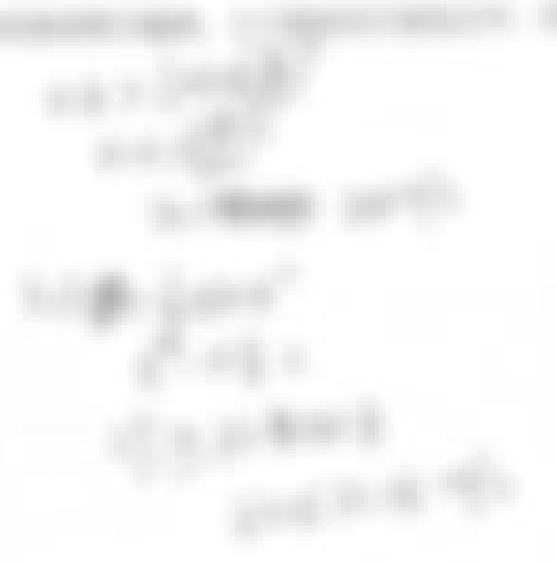
9. $= -\frac{2}{x^3}$

10. $= -\frac{2}{x^3}$

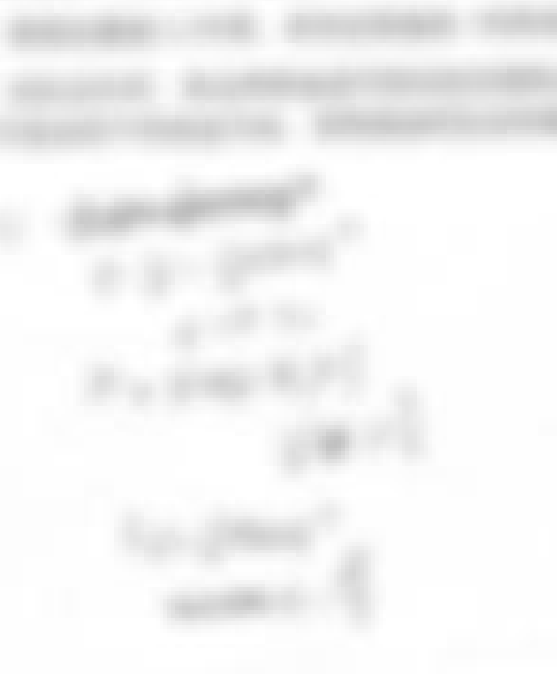
11. $= -\frac{2}{x^3}$



1. The following information is given for the following two questions:
The following information is given for the following two questions:
The following information is given for the following two questions:



2. The following information is given for the following two questions:
The following information is given for the following two questions:
The following information is given for the following two questions:



1. **QUESTION** (10 marks) - The following information relates to the production of a product in a factory. The production process is as follows:

PRODUCTION PROCESS

1. **Raw materials**

2. **Labour**

3. **Overhead costs**

4. **Finished goods**

5. **Selling expenses**

6. **Profit**

7. **Loss**



2. **QUESTION** (10 marks) - The following information relates to the production of a product in a factory. The production process is as follows:

PRODUCTION PROCESS

1. **Raw materials**

2. **Labour**

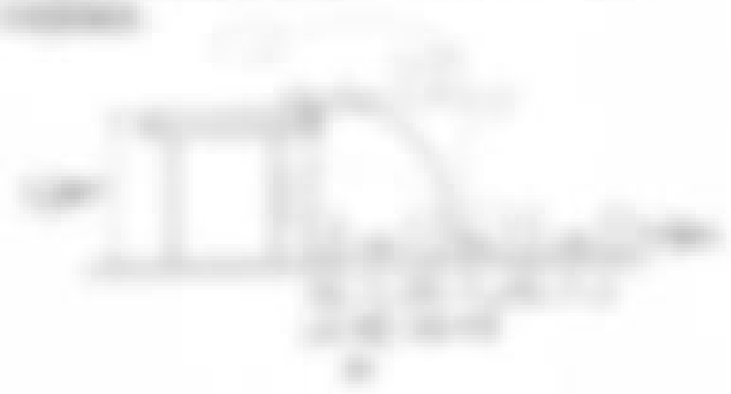
3. **Overhead costs**

4. **Finished goods**

5. **Selling expenses**

6. **Profit**

7. **Loss**



8. **QUESTION**

9. **QUESTION**

10. **QUESTION**

11. **QUESTION**

12. **QUESTION**

1. *Journal of the American Medical Association*, 1997; 277: 1033-1036.

1. Ein Balken der Länge l ist an einem festen Auflager in A eingespannt. Er ist durch eine gleichmäßig verteilte Last q belastet. Berechnen Sie die Auflagerkräfte in A und B .

Gegeben: l , q

4



2. Ein Balken der Länge l ist an einem festen Auflager in A eingespannt. Er ist durch eine gleichmäßig verteilte Last q belastet. Berechnen Sie die Auflagerkräfte in A und B .

Gegeben: l , q



1. **QUESTION:** A rectangular block of material is shown in the figure. The block is subjected to a uniform stress σ acting on its vertical faces. The block is divided into two equal parts by a horizontal plane. The top part is fixed to a rigid support. The bottom part is free to move. The displacement of the bottom part is denoted by δ .



2. **QUESTION:** A rectangular block of material is shown in the figure. The block is subjected to a uniform stress σ acting on its vertical faces. The block is divided into two equal parts by a horizontal plane. The top part is fixed to a rigid support. The bottom part is free to move. The displacement of the bottom part is denoted by δ .

ANSWER: The displacement δ is given by the formula:

$$\delta = \frac{\sigma L}{E}$$

where L is the length of the block, E is the Young's modulus, and σ is the uniform stress.



ANSWER: The displacement δ is given by the formula:

$$\delta = \frac{\sigma L}{E}$$

where L is the length of the block, E is the Young's modulus, and σ is the uniform stress.



1. Introduction (10 minutes)
The purpose of this experiment is to determine the effect of temperature on the rate of reaction between hydrogen peroxide and potassium iodide. The reaction is exothermic and produces oxygen gas and water.

2. Materials (5 minutes)
Hydrogen peroxide solution (10% w/v)
Potassium iodide solution (0.1 M)
Sulfuric acid (1 M)
Sodium thiosulfate solution (0.01 M)
Starch solution (1% w/v)



3. Procedure (15 minutes)
The reaction is carried out in a conical flask. A fixed volume of hydrogen peroxide is added to a fixed volume of potassium iodide solution. The reaction is initiated by the addition of a small amount of sulfuric acid. The time taken for a fixed volume of oxygen to be produced is measured. The experiment is repeated at different temperatures.

4. Results (10 minutes)
The results are recorded in a table. The temperature, volume of oxygen produced, and time taken are noted for each trial. The rate of reaction is calculated as the volume of oxygen produced per unit time.

5. Discussion (10 minutes)
The results show that the rate of reaction increases with temperature. This is because the molecules have more kinetic energy and are more likely to collide successfully. The activation energy of the reaction is estimated to be approximately 50 kJ/mol.



6. Conclusion (5 minutes)
The experiment has shown that the rate of reaction between hydrogen peroxide and potassium iodide increases with temperature. This is consistent with the collision theory of chemical reactions.

1. 1990년대 초반, 우리나라는 경제적으로 어려움을 겪고 있었다. 이 시기에 정부는 경제 활성화를 위한 다양한 정책을 시행하였다. 이 중 가장 대표적인 정책은 무엇이었는가?

2. 다음 중 1990년대 초반 우리나라의 경제 상황을 가장 잘 설명하는 문장은 무엇인가?

- 가. 수출이 크게 증가하였다.
- 나. 외환보유고가 급증하였다.
- 다. 경상수지 흑자가 지속되었다.
- 라. 외환위기에 직면하였다.
- 마. GDP가 크게 성장하였다.
- 바. 인플레이션이 심각하였다.

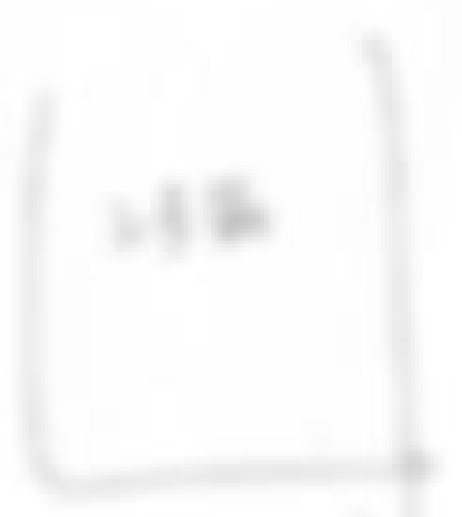


1990년대 초반

3. 1990년대 초반, 우리나라는 경제적으로 어려움을 겪고 있었다. 이 시기에 정부는 경제 활성화를 위한 다양한 정책을 시행하였다. 이 중 가장 대표적인 정책은 무엇이었는가?

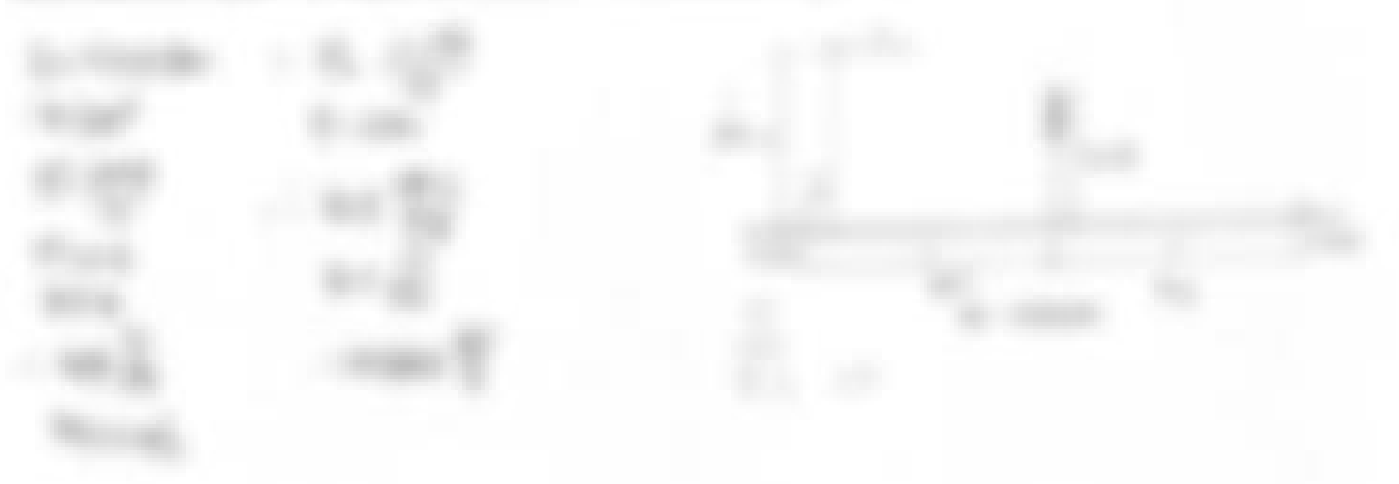
4. 다음 중 1990년대 초반 우리나라의 경제 상황을 가장 잘 설명하는 문장은 무엇인가?

- 가. 수출이 크게 증가하였다.
- 나. 외환보유고가 급증하였다.
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- 라. 외환위기에 직면하였다.
- 마. GDP가 크게 성장하였다.
- 바. 인플레이션이 심각하였다.



1. **QUESTION:** A rectangular plate of width b and height h is subjected to a uniformly distributed load q acting downwards. The plate is supported by a pin support at the left end and a roller support at the right end. Determine the reaction forces at the supports.

SOLUTION: Consider the free-body diagram of the plate.

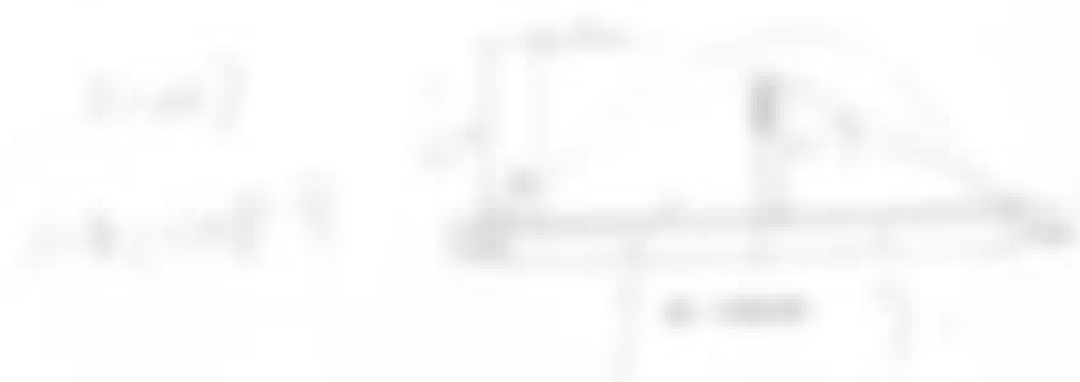


2. **QUESTION:** A rectangular plate of width b and height h is subjected to a uniformly distributed load q acting downwards. The plate is supported by a pin support at the left end and a roller support at the right end. Determine the reaction forces at the supports.

SOLUTION: Consider the free-body diagram of the plate.



1. **QUESTION** (10 marks)
A particle of mass m is projected from the origin O of a Cartesian coordinate system with initial velocity u at an angle θ to the horizontal. The particle moves in a parabolic path and strikes the horizontal axis at a point A . The time taken for the particle to reach A is T . The horizontal distance between O and A is R . The vertical height of the particle at any time t is y . The horizontal distance between O and the particle at any time t is x . The particle is at a height y at two different times t_1 and t_2 . The horizontal distance between the particle at these two times is d . The particle is at a height y at a time t when it is at a horizontal distance x from the origin. The particle is at a height y at a time t when it is at a horizontal distance x from the origin. The particle is at a height y at a time t when it is at a horizontal distance x from the origin.



2. **QUESTION** (10 marks)
A particle of mass m is projected from the origin O of a Cartesian coordinate system with initial velocity u at an angle θ to the horizontal. The particle moves in a parabolic path and strikes the horizontal axis at a point A . The time taken for the particle to reach A is T . The horizontal distance between O and A is R . The vertical height of the particle at any time t is y . The horizontal distance between O and the particle at any time t is x . The particle is at a height y at two different times t_1 and t_2 . The horizontal distance between the particle at these two times is d . The particle is at a height y at a time t when it is at a horizontal distance x from the origin. The particle is at a height y at a time t when it is at a horizontal distance x from the origin. The particle is at a height y at a time t when it is at a horizontal distance x from the origin.



1. **QUESTION** **ANSWER** **QUESTION** **ANSWER**
QUESTION: **QUESTION** **ANSWER** **QUESTION** **ANSWER**
QUESTION: **QUESTION** **ANSWER** **QUESTION** **ANSWER**



2. **QUESTION** **ANSWER** **QUESTION** **ANSWER**
QUESTION: **QUESTION** **ANSWER** **QUESTION** **ANSWER**
QUESTION: **QUESTION** **ANSWER** **QUESTION** **ANSWER**

3. **QUESTION** **ANSWER** **QUESTION** **ANSWER**
QUESTION: **QUESTION** **ANSWER** **QUESTION** **ANSWER**
QUESTION: **QUESTION** **ANSWER** **QUESTION** **ANSWER**



1. **QUESTION** (10 marks) The following diagram shows a beam of length 10 m, supported at its ends by a pin support at A and a roller support at B. A uniformly distributed load of 2 kN/m is applied over the entire length of the beam. Calculate the reactions at A and B.



2. **QUESTION** (10 marks) The following diagram shows a beam of length 10 m, supported at its ends by a pin support at A and a roller support at B. A point load of 10 kN is applied at the midpoint of the beam. Calculate the reactions at A and B.



3. **QUESTION** (10 marks) The following diagram shows a beam of length 10 m, supported at its ends by a pin support at A and a roller support at B. A point load of 10 kN is applied at the midpoint of the beam. Calculate the reactions at A and B.

1. A rectangular block of mass m and height h is placed on a horizontal surface. A horizontal force F is applied to the top edge of the block, perpendicular to the vertical face. The block is on the verge of tipping. Find the magnitude of F .

Weight $= mg$ acting downwards from the center.
Reaction $= R$ acting upwards at the base.
Applied force $= F$ acting horizontally at the top edge.
Distance from pivot to center of mass $= \frac{h}{2}$.
Distance from pivot to point of application of F $= h$.



$\therefore F = \frac{mg}{2}$

2. A rectangular block of mass m and height h is placed on a horizontal surface. A horizontal force F is applied to the top edge of the block, perpendicular to the vertical face. The block is on the verge of tipping. Find the magnitude of F .

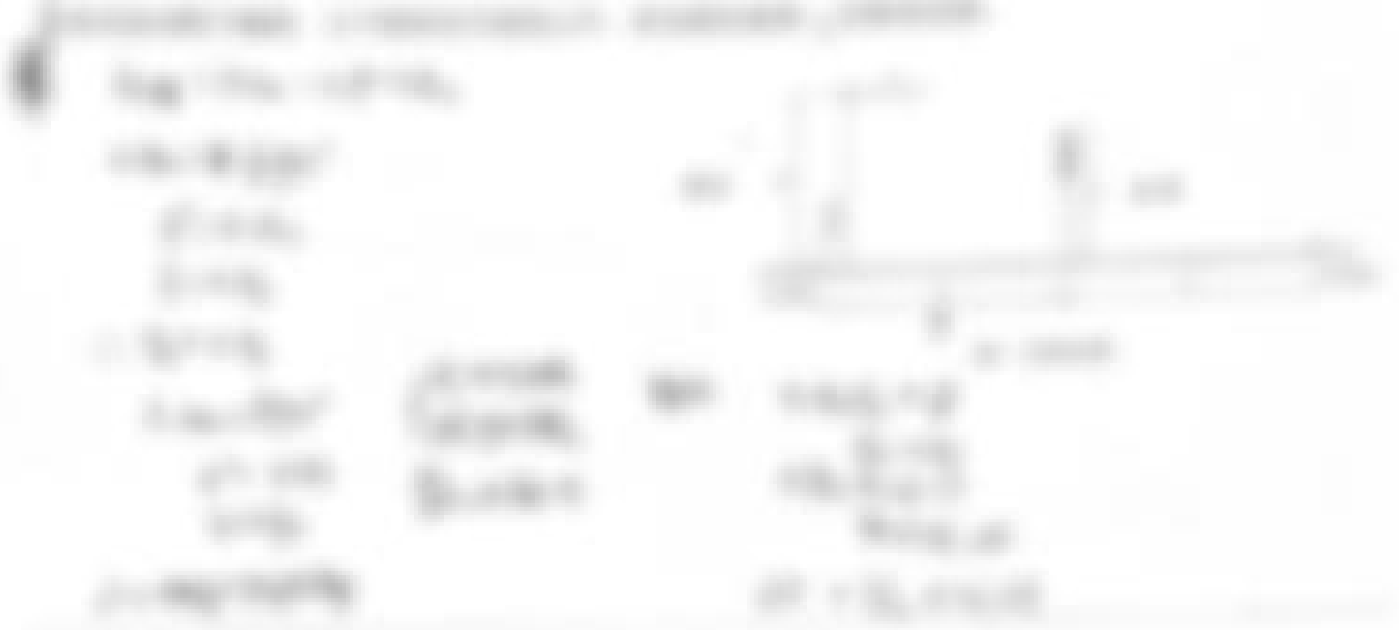
- (a) $F = \frac{mg}{2}$
- (b) $F = \frac{mg}{4}$
- (c) $F = \frac{mg}{3}$
- (d) $F = \frac{mg}{6}$

Weight $= mg$ acting downwards from the center.
Reaction $= R$ acting upwards at the base.
Applied force $= F$ acting horizontally at the top edge.
Distance from pivot to center of mass $= \frac{h}{2}$.
Distance from pivot to point of application of F $= h$.
 $\therefore F = \frac{mg}{2}$



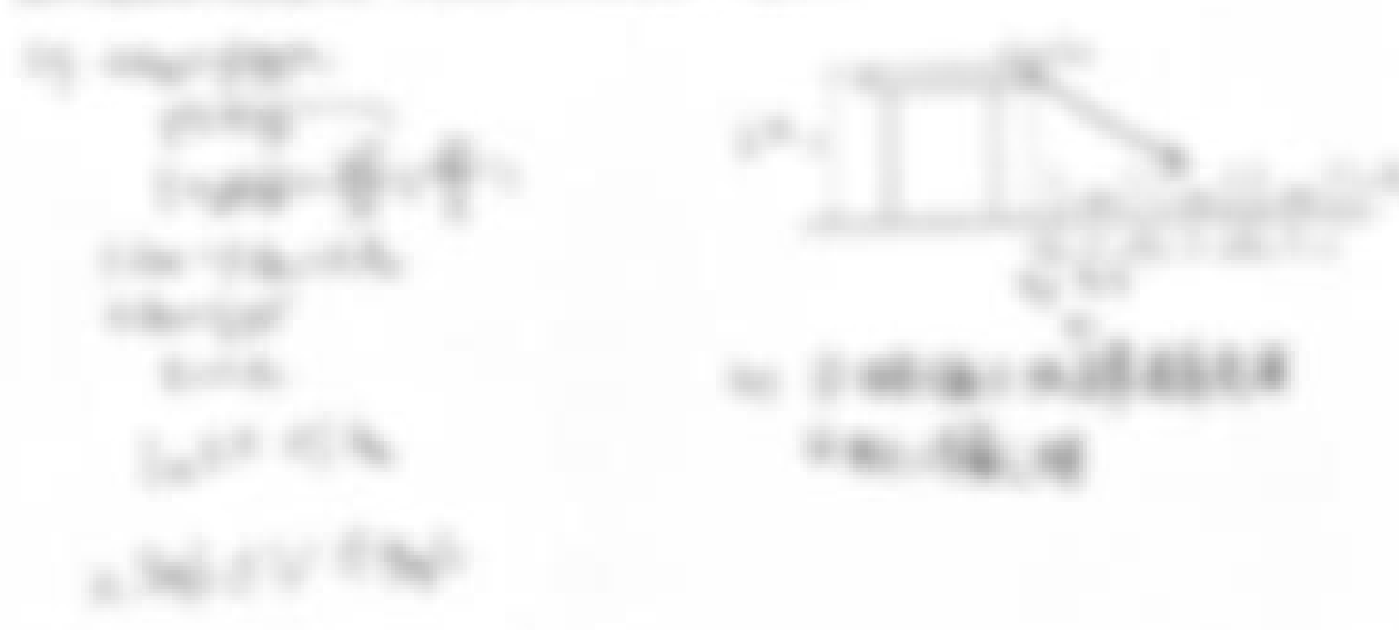
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1. *Handwritten text describing a process or system.*



2. *Handwritten text describing a process or system.*

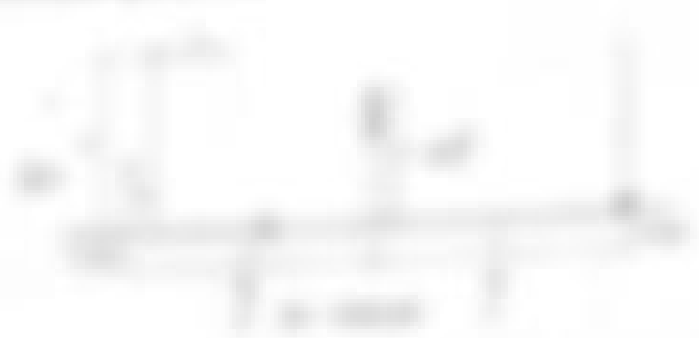
3. *Handwritten text describing a process or system.*



1. *Handwritten text describing a process or system.*

Handwritten text describing a process or system.

Handwritten notes or calculations.



2. *Handwritten text describing a process or system.*

Handwritten text describing a process or system.

3. *Handwritten text describing a process or system.*

Handwritten notes or calculations.

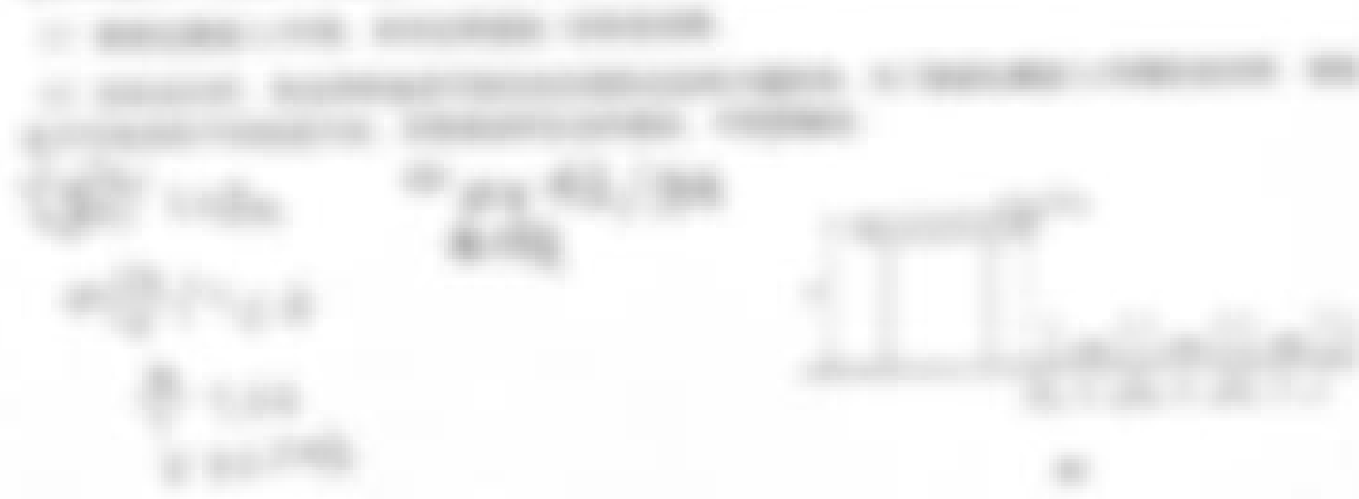
Handwritten notes or calculations.



1. **QUESTION** (10 marks)
A rectangular block of material is shown in the figure below. The block is subjected to a uniform stress σ acting on the vertical faces. The block is divided into two equal parts by a vertical plane. The top part is fixed to a rigid wall, and the bottom part is free to move. The block is shown in the figure below.



2. **ANSWER** (10 marks)
The block is subjected to a uniform stress σ acting on the vertical faces. The block is divided into two equal parts by a vertical plane. The top part is fixed to a rigid wall, and the bottom part is free to move. The block is shown in the figure below.



1. Handwritten title
2. Handwritten subtitle
3. Handwritten text
4. Handwritten text



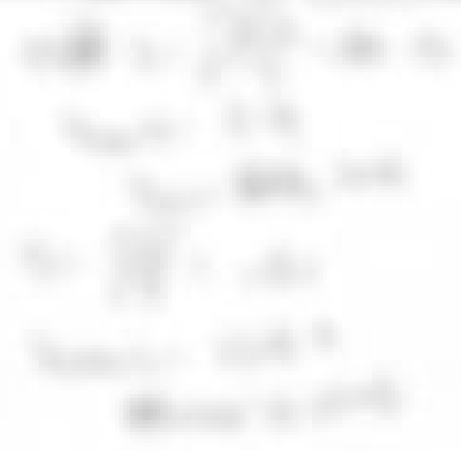
1. Handwritten title
2. Handwritten subtitle
3. Handwritten text
4. Handwritten text



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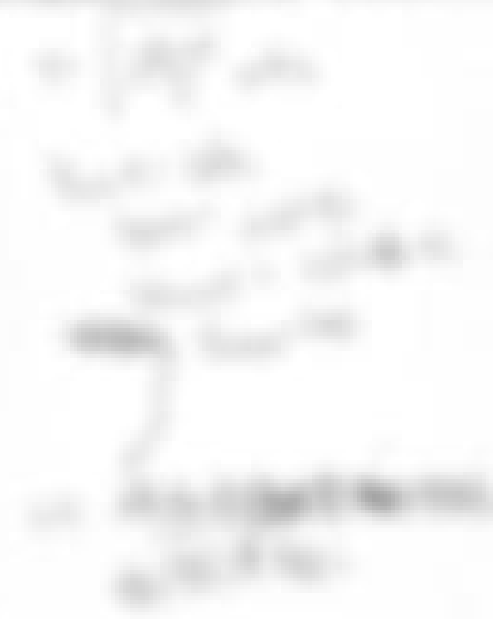
1. Handwritten text describing the first part of the experiment, mentioning the setup and the initial conditions.

Handwritten text describing the second part of the experiment, mentioning the setup and the initial conditions.



2. Handwritten text describing the second part of the experiment, mentioning the setup and the initial conditions.

3. Handwritten text describing the third part of the experiment, mentioning the setup and the initial conditions.



1. **QUESTION:** A rectangular block of material is shown in the figure. The block is subjected to a uniform stress σ acting on the vertical faces. The block is divided into two equal parts by a vertical plane. The top part is fixed to a wall, and the bottom part is free to move. The block is shown in the figure.

ANSWER: The block is shown in the figure.

Q. A rectangular block of material is shown in the figure. The block is subjected to a uniform stress σ acting on the vertical faces. The block is divided into two equal parts by a vertical plane. The top part is fixed to a wall, and the bottom part is free to move. The block is shown in the figure.



2. **QUESTION:** A rectangular block of material is shown in the figure. The block is subjected to a uniform stress σ acting on the vertical faces. The block is divided into two equal parts by a vertical plane. The top part is fixed to a wall, and the bottom part is free to move. The block is shown in the figure.

ANSWER: The block is shown in the figure.

3. **QUESTION:** A rectangular block of material is shown in the figure. The block is subjected to a uniform stress σ acting on the vertical faces. The block is divided into two equal parts by a vertical plane. The top part is fixed to a wall, and the bottom part is free to move. The block is shown in the figure.

Q. A rectangular block of material is shown in the figure. The block is subjected to a uniform stress σ acting on the vertical faces. The block is divided into two equal parts by a vertical plane. The top part is fixed to a wall, and the bottom part is free to move. The block is shown in the figure.



A. A rectangular block of material is shown in the figure. The block is subjected to a uniform stress σ acting on the vertical faces. The block is divided into two equal parts by a vertical plane. The top part is fixed to a wall, and the bottom part is free to move. The block is shown in the figure.

Handwritten signature

1. The following information is for the purpose of the project. It is not to be used for any other purpose. It is the property of the company and should be kept confidential.

2. The following information is for the purpose of the project. It is not to be used for any other purpose. It is the property of the company and should be kept confidential.



3. The following information is for the purpose of the project. It is not to be used for any other purpose. It is the property of the company and should be kept confidential.

4. The following information is for the purpose of the project. It is not to be used for any other purpose. It is the property of the company and should be kept confidential.

5. The following information is for the purpose of the project. It is not to be used for any other purpose. It is the property of the company and should be kept confidential.

6. The following information is for the purpose of the project. It is not to be used for any other purpose. It is the property of the company and should be kept confidential.

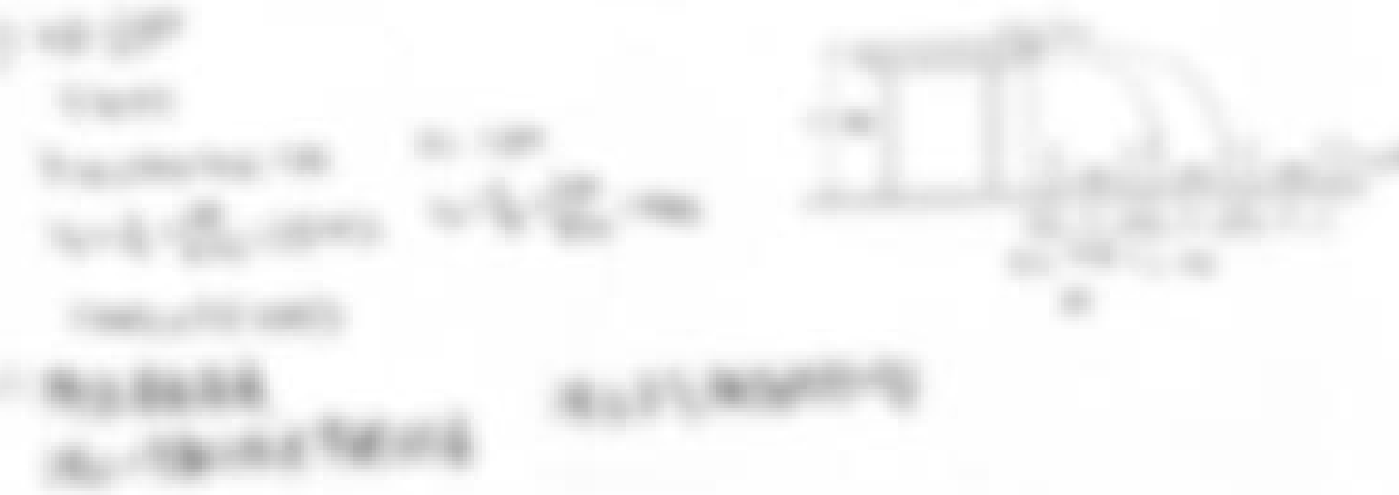


1. The following information is given for the company for the year ended 31st March 2014:

	£	£
Revenue	1000	
Cost of sales	(400)	
Gross profit	600	
Less: Administrative expenses	(200)	
Less: Selling expenses	(100)	
Profit before tax	300	
Less: Tax	(100)	
Profit after tax	200	

2. The following information is given for the company for the year ended 31st March 2014:

- (a) Revenue: 1000
- (b) Cost of sales: 400
- (c) Administrative expenses: 200
- (d) Selling expenses: 100
- (e) Profit before tax: 300
- (f) Tax: 100
- (g) Profit after tax: 200



1. *[Faint text]*

[Faint text]

[Faint handwritten notes]

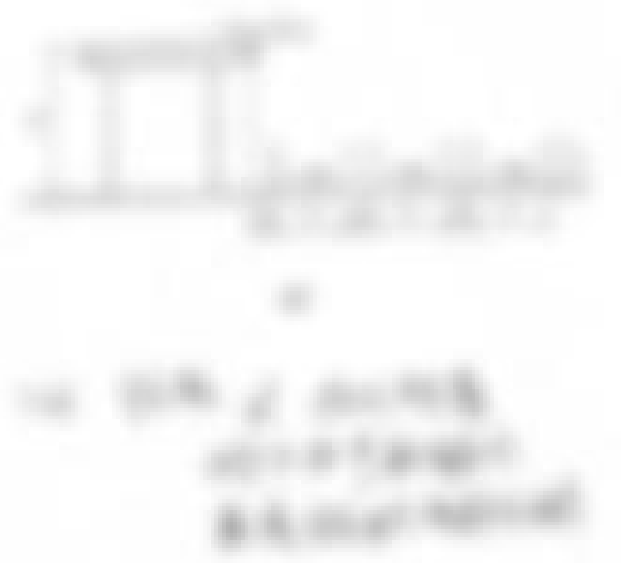


2. *[Faint text]*

[Faint text]

[Faint text]

[Faint handwritten notes]



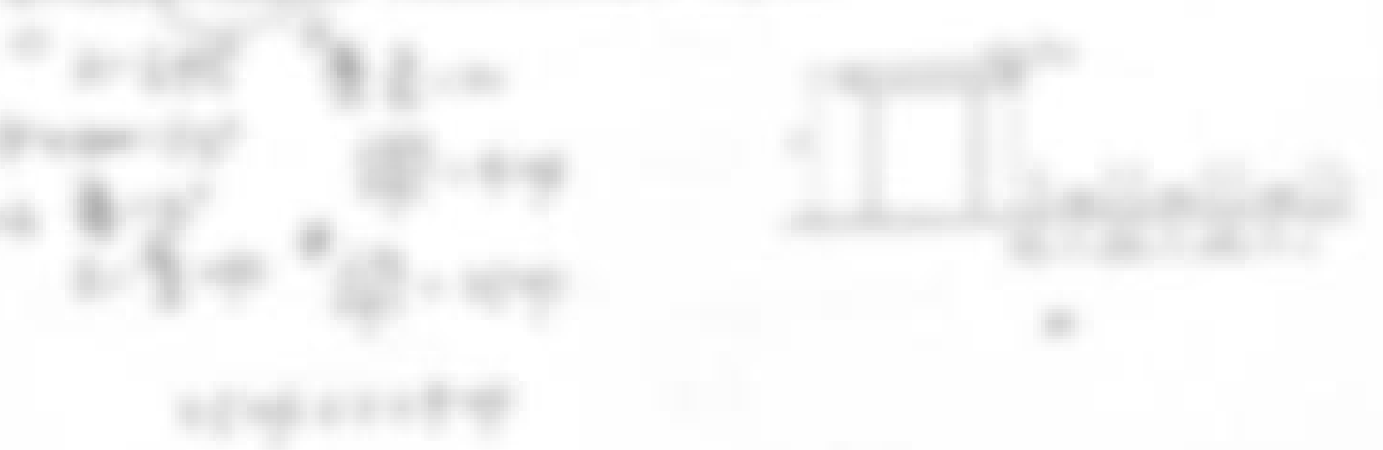
1. *Journal of the American Medical Association*, 1998; 279: 1001-1005.

1. The first step in the process of the scientific method is to make an observation or ask a question. This leads to the second step, which is to do background research.



2. The second step is to do background research. This involves looking up information about the topic to see what is already known and to help form a hypothesis.

3. The third step is to make a hypothesis. This is an educated guess about what the results of the experiment will be.



4. The fourth step is to conduct an experiment. This is where you test your hypothesis by collecting data.

1. Introduction (background information)
The purpose of this study is to investigate the effects of...
on...
The study was conducted in...
The results of the study are as follows:

Methodology (research design, participants, procedures)

Results
The results of the study are presented in the following tables and figures.
Table 1: Descriptive statistics
Table 2: ANOVA results
Figure 1: Line graph showing the relationship between...
The results indicate that...
There was a significant difference between...
The effect size was...
The results are consistent with previous research...



Figure 1: Line graph showing the relationship between...
(caption text)

2. Discussion (interpretation of results, implications, limitations)
The findings of this study suggest that...
These results have important implications for...
The study has several limitations, including...
Future research should focus on...
The study was limited to...
The results may not be generalizable to...
The study was conducted in a controlled environment...

3. Conclusion (summary of findings)

4. References (list of sources cited)

Appendix
Appendix A: Raw data
Appendix B: Interview transcripts
Appendix C: Questionnaire results
Appendix D: Ethical approval forms
Appendix E: Informed consent forms
Appendix F: Research protocol
Appendix G: Statistical analysis output
Appendix H: Researcher's CV
Appendix I: Funding statement
Appendix J: Acknowledgements

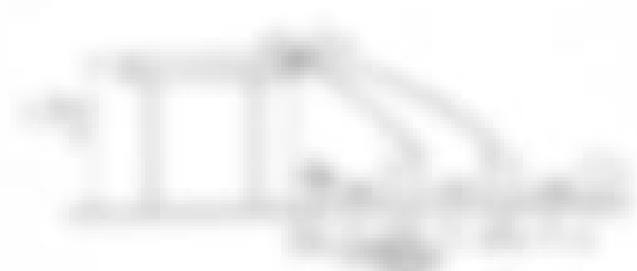


Figure 2: Bar chart showing the relationship between...
(caption text)

Figure 3: Scatter plot showing the relationship between...
(caption text)

1. Diagramm des Versuchsaufbaus: Ein Zylinder mit einem Volumen V ist mit einem Gas gefüllt. Die Zylinderwand ist aus einem Material, das eine Wärmeleitfähigkeit λ hat. Die Zylinderlänge ist l . Die Zylinderwanddicke ist δ . Die Zylinderwand ist an beiden Enden mit einem Material, das eine Wärmeleitfähigkeit λ_1 hat. Die Zylinderwand ist an beiden Enden mit einem Material, das eine Wärmeleitfähigkeit λ_2 hat.

2. Diagramm des Versuchsaufbaus:

3. Diagramm des Versuchsaufbaus:

4. Diagramm des Versuchsaufbaus:

5. Diagramm des Versuchsaufbaus:

6. Diagramm des Versuchsaufbaus:



7. Diagramm des Versuchsaufbaus: Ein Zylinder mit einem Volumen V ist mit einem Gas gefüllt. Die Zylinderwand ist aus einem Material, das eine Wärmeleitfähigkeit λ hat. Die Zylinderlänge ist l . Die Zylinderwanddicke ist δ . Die Zylinderwand ist an beiden Enden mit einem Material, das eine Wärmeleitfähigkeit λ_1 hat. Die Zylinderwand ist an beiden Enden mit einem Material, das eine Wärmeleitfähigkeit λ_2 hat.

8. Diagramm des Versuchsaufbaus:

9. Diagramm des Versuchsaufbaus: Ein Zylinder mit einem Volumen V ist mit einem Gas gefüllt. Die Zylinderwand ist aus einem Material, das eine Wärmeleitfähigkeit λ hat. Die Zylinderlänge ist l . Die Zylinderwanddicke ist δ . Die Zylinderwand ist an beiden Enden mit einem Material, das eine Wärmeleitfähigkeit λ_1 hat. Die Zylinderwand ist an beiden Enden mit einem Material, das eine Wärmeleitfähigkeit λ_2 hat.

10. Diagramm des Versuchsaufbaus:

11. Diagramm des Versuchsaufbaus:

12. Diagramm des Versuchsaufbaus:

13. Diagramm des Versuchsaufbaus:

14. Diagramm des Versuchsaufbaus:

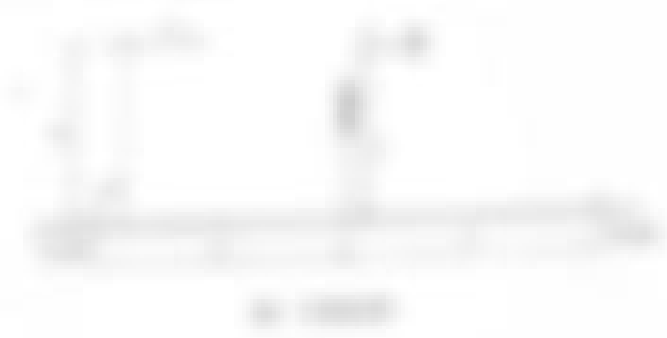
15. Diagramm des Versuchsaufbaus:



1. A rectangular block of mass m and height h is placed on a horizontal surface. A horizontal force F is applied to the block at a height y from the bottom. The block is on the verge of tipping over the right edge. Find the magnitude of F in terms of m , h , and y .

2. A rectangular block of mass m and height h is placed on a horizontal surface. A horizontal force F is applied to the block at a height y from the bottom. The block is on the verge of tipping over the right edge. Find the magnitude of F in terms of m , h , and y .

$$\begin{aligned} \sum \tau = 0 \\ m \cdot g \cdot \frac{h}{2} &= F \cdot y \\ F &= \frac{m \cdot g \cdot h}{2y} \end{aligned}$$



3. A rectangular block of mass m and height h is placed on a horizontal surface. A horizontal force F is applied to the block at a height y from the bottom. The block is on the verge of tipping over the right edge. Find the magnitude of F in terms of m , h , and y .

4. A rectangular block of mass m and height h is placed on a horizontal surface. A horizontal force F is applied to the block at a height y from the bottom. The block is on the verge of tipping over the right edge. Find the magnitude of F in terms of m , h , and y .

$$\begin{aligned} \text{A} \quad \textcircled{D} \quad \frac{mgh}{2y} \\ \text{B} \quad \textcircled{A} \quad \frac{mgh}{y} \end{aligned}$$



1. $\frac{1}{x^2} = x^{-2}$ $\frac{d}{dx} x^{-2} = -2x^{-3} = -\frac{2}{x^3}$
2. $\frac{1}{x^3} = x^{-3}$ $\frac{d}{dx} x^{-3} = -3x^{-4} = -\frac{3}{x^4}$
3. $\frac{1}{x^4} = x^{-4}$ $\frac{d}{dx} x^{-4} = -4x^{-5} = -\frac{4}{x^5}$

4. $\frac{1}{x^5} = x^{-5}$ $\frac{d}{dx} x^{-5} = -5x^{-6} = -\frac{5}{x^6}$

$$\begin{aligned} \frac{d}{dx} x^{-2} &= -2x^{-3} \\ \frac{d}{dx} x^{-3} &= -3x^{-4} \\ \frac{d}{dx} x^{-4} &= -4x^{-5} \\ \frac{d}{dx} x^{-5} &= -5x^{-6} \\ \frac{d}{dx} x^{-6} &= -6x^{-7} \\ \frac{d}{dx} x^{-7} &= -7x^{-8} \end{aligned}$$



$$\begin{aligned} \frac{d}{dx} \frac{1}{x^2} &= \frac{d}{dx} x^{-2} = -2x^{-3} = -\frac{2}{x^3} \\ \frac{d}{dx} \frac{1}{x^3} &= \frac{d}{dx} x^{-3} = -3x^{-4} = -\frac{3}{x^4} \\ \frac{d}{dx} \frac{1}{x^4} &= \frac{d}{dx} x^{-4} = -4x^{-5} = -\frac{4}{x^5} \\ \frac{d}{dx} \frac{1}{x^5} &= \frac{d}{dx} x^{-5} = -5x^{-6} = -\frac{5}{x^6} \end{aligned}$$

5. $\frac{1}{x^6} = x^{-6}$ $\frac{d}{dx} x^{-6} = -6x^{-7} = -\frac{6}{x^7}$
6. $\frac{1}{x^7} = x^{-7}$ $\frac{d}{dx} x^{-7} = -7x^{-8} = -\frac{7}{x^8}$
7. $\frac{1}{x^8} = x^{-8}$ $\frac{d}{dx} x^{-8} = -8x^{-9} = -\frac{8}{x^9}$

8. $\frac{1}{x^9} = x^{-9}$ $\frac{d}{dx} x^{-9} = -9x^{-10} = -\frac{9}{x^{10}}$

9. $\frac{1}{x^{10}} = x^{-10}$ $\frac{d}{dx} x^{-10} = -10x^{-11} = -\frac{10}{x^{11}}$

$$\begin{aligned} \frac{d}{dx} x^{-6} &= -6x^{-7} \\ \frac{d}{dx} x^{-7} &= -7x^{-8} \\ \frac{d}{dx} x^{-8} &= -8x^{-9} \\ \frac{d}{dx} x^{-9} &= -9x^{-10} \\ \frac{d}{dx} x^{-10} &= -10x^{-11} \end{aligned}$$



10. $\frac{1}{x^{11}} = x^{-11}$ $\frac{d}{dx} x^{-11} = -11x^{-12} = -\frac{11}{x^{12}}$
11. $\frac{1}{x^{12}} = x^{-12}$ $\frac{d}{dx} x^{-12} = -12x^{-13} = -\frac{12}{x^{13}}$
12. $\frac{1}{x^{13}} = x^{-13}$ $\frac{d}{dx} x^{-13} = -13x^{-14} = -\frac{13}{x^{14}}$

1. **QUESTION:** A beam of length 10 m is supported at its left end by a pin support and at its right end by a roller support. A uniformly distributed load of 2 kN/m is applied over the entire length of the beam. Determine the reaction forces at the supports.

SOLUTION: Draw a free-body diagram of the beam.

Let R_L be the reaction force at the left support and R_R be the reaction force at the right support. The uniformly distributed load can be replaced by a single point load of 20 kN acting at the center of the beam, i.e., 5 m from the left support.



1. **ANSWER:** $R_L = 10 \text{ kN}$, $R_R = 10 \text{ kN}$

2. **QUESTION:** A beam of length 10 m is supported at its left end by a pin support and at its right end by a roller support. A point load of 10 kN is applied at a distance of 3 m from the left support. Determine the reaction forces at the supports.

SOLUTION: Draw a free-body diagram of the beam.

Let R_L be the reaction force at the left support and R_R be the reaction force at the right support. The point load of 10 kN acts downwards at $x = 3 \text{ m}$ from the left support.

Let R_L be the reaction force at the left support and R_R be the reaction force at the right support. The point load of 10 kN acts downwards at $x = 3 \text{ m}$ from the left support.



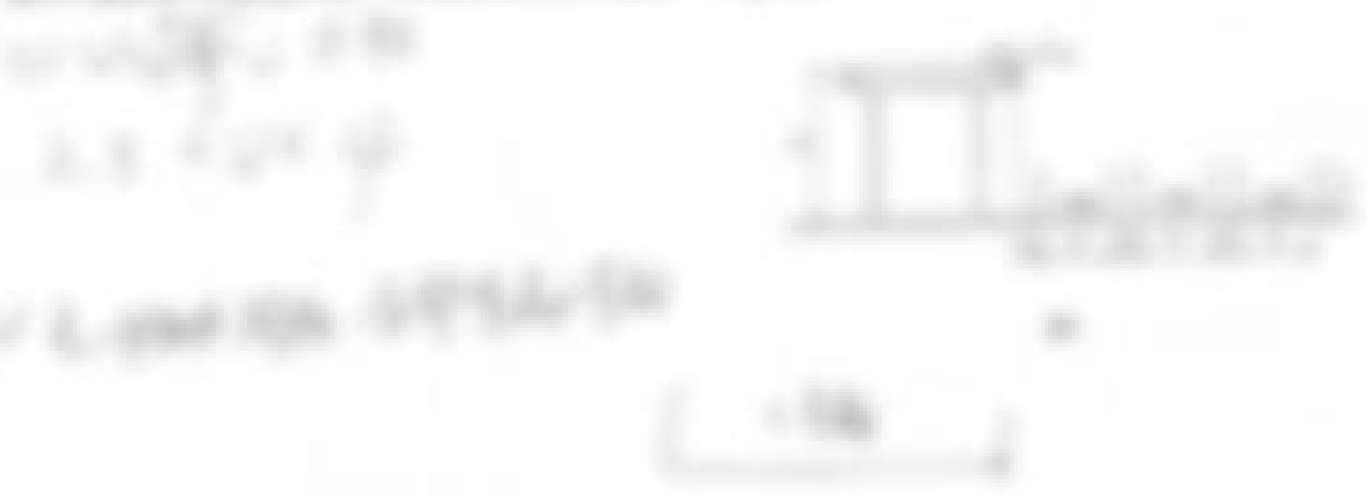
Let R_L be the reaction force at the left support and R_R be the reaction force at the right support. The point load of 10 kN acts downwards at $x = 3 \text{ m}$ from the left support.

1. The following information is for your information only. It is not to be used for any other purpose. It is not to be used for any other purpose.



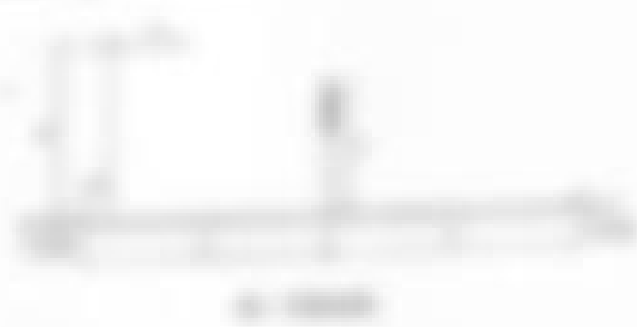
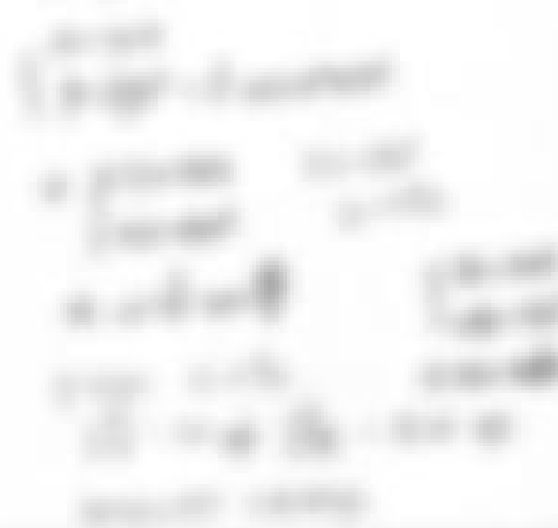
2. The following information is for your information only. It is not to be used for any other purpose. It is not to be used for any other purpose.

3. The following information is for your information only. It is not to be used for any other purpose. It is not to be used for any other purpose.



1. **QUESTION:** A beam of light is incident on a surface at an angle of 30 degrees to the normal. What is the angle of reflection?

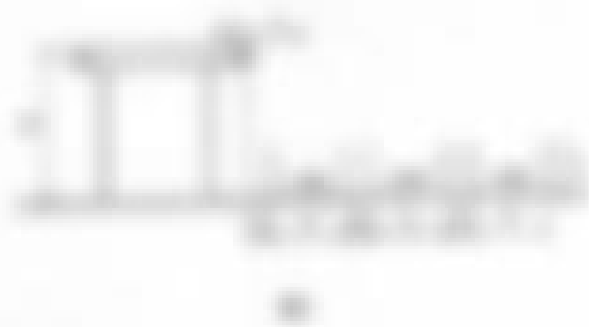
ANSWER: The angle of reflection is 30 degrees.



2. **QUESTION:** A beam of light is incident on a surface at an angle of 45 degrees to the normal. What is the angle of reflection?

ANSWER: The angle of reflection is 45 degrees.

3. **QUESTION:** A beam of light is incident on a surface at an angle of 60 degrees to the normal. What is the angle of reflection?



1. The following information is for the purpose of providing a summary of the information contained in the report. It is not intended to be a substitute for the full report.

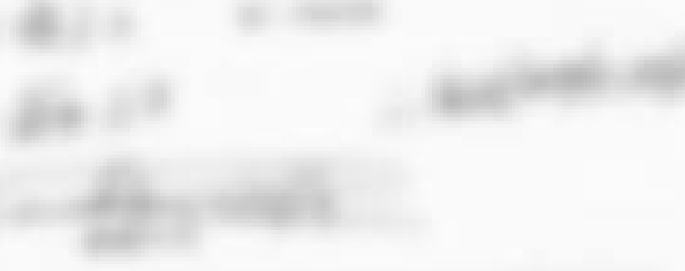


2. The following information is for the purpose of providing a summary of the information contained in the report. It is not intended to be a substitute for the full report.

3. The following information is for the purpose of providing a summary of the information contained in the report. It is not intended to be a substitute for the full report.



1. Geometrische Optik
Lichtstrahlen verlaufen in geraden Linien.
Lichtstrahlen ändern ihre Richtung an Grenzflächen.
Lichtstrahlen werden durch Linsen gebrochen.
Lichtstrahlen werden durch Spiegel reflektiert.
Lichtstrahlen werden durch Gitter gebeugt.
Lichtstrahlen werden durch Kristalle gebrochen.
Lichtstrahlen werden durch Polarisatoren polarisiert.
Lichtstrahlen werden durch Interferenz überlagert.
Lichtstrahlen werden durch Diffraction gebeugt.
Lichtstrahlen werden durch Streuung verstreut.



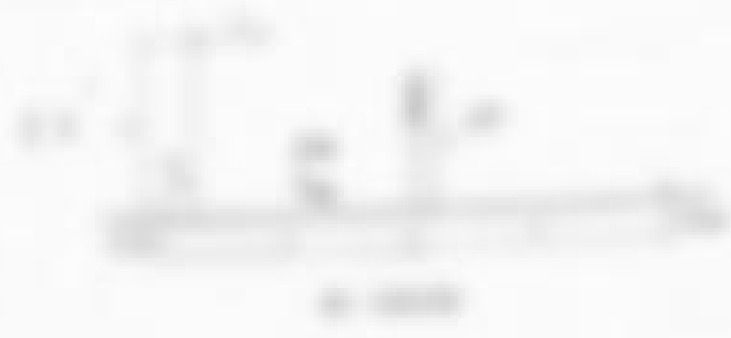
2. Physikalische Optik
Lichtstrahlen verhalten sich wie Wellen.
Lichtstrahlen werden durch Interferenz überlagert.
Lichtstrahlen werden durch Diffraction gebeugt.
Lichtstrahlen werden durch Streuung verstreut.
Lichtstrahlen werden durch Polarisation polarisiert.
Lichtstrahlen werden durch Dispersion zerlegt.
Lichtstrahlen werden durch Reflexion reflektiert.
Lichtstrahlen werden durch Brechung gebrochen.
Lichtstrahlen werden durch Absorption absorbiert.
Lichtstrahlen werden durch Emission emittiert.



1. *[Faint text]*

[Faint text]

[Faint text]



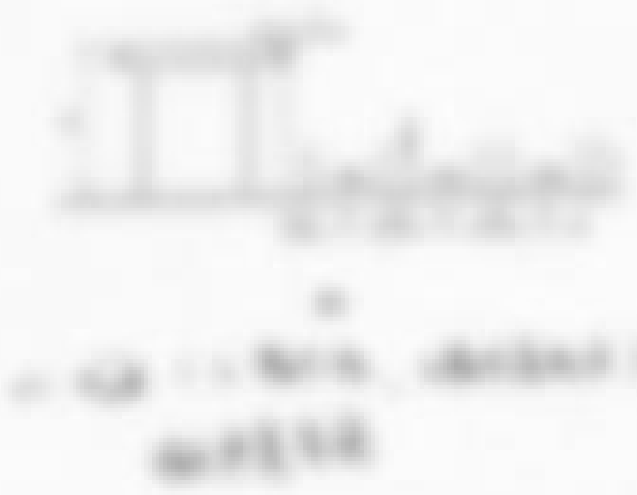
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2. *[Faint text]*

[Faint text]

3. *[Faint text]*

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© 2000 Blackwell Science Ltd, *Journal of Internal Medicine* 247: 369–375

1. *[Faint text]*

[Faint text]

$\frac{1}{2} = \frac{1}{2} - \frac{1}{2}$
 $\frac{1}{2} = \frac{1}{2}$
 $\frac{1}{2} = \frac{1}{2} - \frac{1}{2}$
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 $\frac{1}{2} = \frac{1}{2}$
 $\frac{1}{2} = \frac{1}{2} - \frac{1}{2}$



[Faint text]

2. *[Faint text]*

[Faint text]

3. *[Faint text]*

$\frac{1}{2} = \frac{1}{2} - \frac{1}{2}$
 $\frac{1}{2} = \frac{1}{2}$



[Faint text]

[Faint text]



[Faint text]

1. The following information is available for the year ended 31/12/2019:

Required: Calculate the following ratios.

- (a) Current ratio
- (b) Acid test ratio
- (c) Gross profit ratio
- (d) Net profit ratio



- (e) Operating ratio
- (f) Operating leverage
- (g) Operating cycle
- (h) Inventory turnover ratio

2. The following information is available for the year ended 31/12/2019:

Required: Calculate the following ratios.

- (a) Current ratio
- (b) Acid test ratio
- (c) Gross profit ratio
- (d) Net profit ratio
- (e) Operating ratio
- (f) Operating leverage
- (g) Operating cycle
- (h) Inventory turnover ratio



- (i) Operating ratio
- (j) Operating leverage
- (k) Operating cycle
- (l) Inventory turnover ratio

1. Handwritten title
2. Handwritten address
3. Handwritten date

Handwritten subject line

1. Handwritten name
2. Handwritten address

1. Handwritten name
2. Handwritten address
3. Handwritten date
4. Handwritten subject

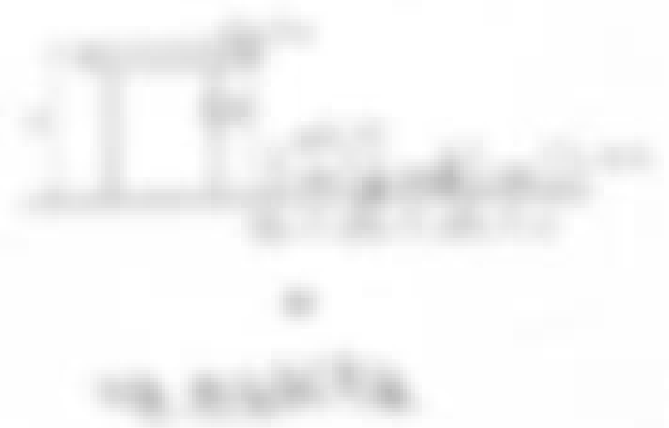


1. Handwritten name (in parentheses)
2. Handwritten address
3. Handwritten date
4. Handwritten subject

1. Handwritten title
2. Handwritten address
3. Handwritten date
4. Handwritten subject

1. Handwritten title
2. Handwritten address
3. Handwritten date
4. Handwritten subject

1. Handwritten name
2. Handwritten address
3. Handwritten date
4. Handwritten subject



1. Handwritten name
2. Handwritten address
3. Handwritten date
4. Handwritten subject

1. *[Faint text]*

[Faint text]

[Faint text]



2. *[Faint text]*

[Faint text]

[Faint text]



[Faint text]



The reflection coefficient R is defined as the ratio of the reflected power to the incident power. For a dielectric slab, the reflection coefficient is given by:

$$R = \frac{1 - \cos(2\theta_i)}{1 + \cos(2\theta_i)}$$

where θ_i is the angle of incidence. The reflection coefficient is a function of the angle of incidence and the dielectric constant of the slab.



1. The following information is available for the year ended 31st December 2019:

Particulars Amount (£)

Revenue 1000

Cost of sales 600

Depreciation 50

Interest 20

Profit before tax 230



2. The following information is available for the year ended 31st December 2019:

Particulars Amount (£)

Revenue 1000
Cost of sales 600
Depreciation 50
Interest 20
Profit before tax 230

Revenue 1000

Cost of sales 600



Profit before tax 230

1. The following information was obtained from the records of the Department of Health and Human Services, Office of the Assistant Secretary for Health, regarding the activities of the National Health and Medical Research Council (NH&MRC) during the period 1970-1971.

2. The following information was obtained from the records of the Department of Health and Human Services, Office of the Assistant Secretary for Health, regarding the activities of the National Health and Medical Research Council (NH&MRC) during the period 1970-1971.



3. The following information was obtained from the records of the Department of Health and Human Services, Office of the Assistant Secretary for Health, regarding the activities of the National Health and Medical Research Council (NH&MRC) during the period 1970-1971.

4. The following information was obtained from the records of the Department of Health and Human Services, Office of the Assistant Secretary for Health, regarding the activities of the National Health and Medical Research Council (NH&MRC) during the period 1970-1971.



1. The following information is given for the structure shown in the figure. The structure is a frame with a horizontal beam of length 10 m and a vertical column of height 4 m. The beam is supported by a pin support at the left end and a roller support at the right end. The column is supported by a pin support at the base. A uniformly distributed load of 2 kN/m is applied vertically downwards over the entire length of the beam. A point load of 10 kN is applied vertically downwards at the free end of the beam. The angle of inclination of the beam with the horizontal is 30°.

Calculate the reactions at the supports.



2. The following information is given for the structure shown in the figure. The structure is a frame with a horizontal beam of length 10 m and a vertical column of height 4 m. The beam is supported by a pin support at the left end and a roller support at the right end. The column is supported by a pin support at the base. A uniformly distributed load of 2 kN/m is applied vertically downwards over the entire length of the beam. A point load of 10 kN is applied vertically downwards at the free end of the beam. The angle of inclination of the beam with the horizontal is 30°.

- (a) Calculate the reactions at the supports.
- (b) Calculate the shear force and bending moment at the free end of the beam.

